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A large neighborhood search approach to the vehicle routing problem with delivery options

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ABSTRACT

To reduce delivery failures in last mile delivery, several types of delivery options have been proposed in the past twenty years. Still, customer satisfaction is a challenge because a single location is chosen independently of the time at which the customer's order will be delivered. In addition, delivery at shared locations such as lockers and shops also experience failure due to capacity or opening-time issues at the moment of delivery. To address this issue and foster consolidation at shared delivery locations, we investigate the case where a customer can specify several delivery options together with preference levels and time windows. We define, in this article, the Vehicle Routing Problem with Delivery Options, which integrates several types of delivery locations. It consists of designing a set of routes for a fleet of vehicles that deliver to each customer at one of his/her options during the corresponding time window. These routes should respect capacities at shared locations such as lockers and minimum service level requirements, while minimizing the total routing costs. This problem is solved with a large neighborhood search in which a set partitioning problem is periodically used to reassemble routes. Specific ruin and recreate operators are proposed and combined with numerous operators from the literature. A thorough experimental study was carried out to determine a subset of efficient and complementary operators. The proposed method outperforms existing algorithms from the literature on particular cases of the problem under consideration, such as the vehicle routing problem with roaming delivery locations and the vehicle routing problem with home and roaming delivery locations. New instances are generated and used both to serve as a benchmark and to propose some managerial insight into the vehicle routing problem with alternative delivery options.

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1. Introduction

The revenue of online stores in France was 92.6 billion euros in 2018, after an average growth of 13% per year during the last four years (Moyou, 2019). With this growth of e-commerce, an increasing number of parcels have to be delivered each day. Consequently, several possibilities have been developed so that distribution can be faster and cheaper. According to Morganti et al. (2014), in 2012, 90% of the French and German population were within 10 minutes of a locker or a pick-up point (often shops or post offices), and the number of such locations grew by 33% between 2008 and 2012. Furthermore, trunk deliveries were recently tested on an industrial scale (McFarland, 2018).

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The customer is generally required to choose one delivery location. Nevertheless, people move around during the day, for example, to go to work or take children to school. This is the major cause of no-show in attended deliveries. According to Allen et al. (2016), in the UK, 14% of all deliveries fail. In 2014, in the UK, the cost of these failed deliveries has been estimated at £771 million. Consequently, it is very likely that a purchaser would like to choose between several delivery options depending on the time of delivery.

The objective of carriers is to deliver all their parcels at a minimum cost. They also want to maintain a good quality of service, but the time window width has a great impact on delivery costs. Nevertheless, they are essential because 80% of the parcels delivered in the UK do not fit into the letterbox (Allen et al., 2016) and nobody likes to wait all afternoon at home for a parcel (Agatz et al., 2008). Accordingly, new strategies are developed to increase the number of successful deliveries (Florio et al., 2018).

This paper discusses the Vehicle Routing Problem with Delivery Options (VRPDO). It is an extension of both the Vehicle Routing Problem with Time Windows (VRPTW) (Savelsbergh, 1985) and the Generalized Vehicle Routing Problem (GVRP) (Ghiani and Improta, 2000). In the VRPDO, each customer can choose to have a package delivered through several delivery options. For example, a given parcel can be delivered at office during work hours, at home in the evening or in a locker at any time.

From the standpoint of the carrier, these delivery options can be of two natures. They can take place in individual delivery locations, like a home or a car trunk. This means that only one customer can be delivered in that location. Otherwise, deliveries can take place at shared delivery locations (SDL), like lockers or pick-up points. This means that several parcels can be left at the same location, while possibly satisfying some capacity constraints.

The contribution of this paper is two-fold. First, from the managerial point of view, a new variant of the Vehicle Routing Problem (VRP) is introduced: the VRPDO combines delivery options to reduce delivery costs while guaranteeing a higher quality of service. From the methodological point of view, a Large Neighborhood Search (LNS) metaheuristics that embeds many recent and new ideas is proposed. In particular, this LNS includes many operators from the literature and compares them rigorously.

The paper is structured as follows: In Section 2, the VRPDO is described and a mathematical model is proposed. Section 3 exposes the literature on related problems. Section 4 details the LNS developed to solve the VRPDO. In addition, our methodology to configure the algorithm is explained. Finally, Section 5 develops experiments, first to validate the method and second to draw managerial insights from randomly generated instances of the VRPDO.

2. The vehicle routing problem with delivery options

This section details all components of the VRPDO (Section 2.1) and gives a mathematical model (Section 2.2).

2.1. Problem settings

The VRPDO is an operational optimization problem defined on a short time horizon, typically one day. In this problem, each customer can choose several *delivery options*. In the following, for the sake of conciseness, we will refer to them only as *options*.

An option is a tuple associated with a unique customer, composed of *a location, a preference level* and *a service duration*, as follows:

- A *location* has a geographical address and a time window during which goods can be delivered. For each location, a fixed preparation time represents the time needed to park a vehicle and access the delivery point. Two types of locations are considered: individual delivery locations and SDLs. An individual delivery location is typically a personal address. Only one option can be associated with this location. It generally has a tight time window. An SDL is typically a pickup point or a locker. Several options can be associated with this location. An SDL can be given a capacity, which is defined as the maximum number of packages which can be delivered at this location during the time horizon. For the sake of conciseness, SDLs and individual delivery locations may be called “shared locations” and “individual locations”, respectively.
- The *preference level* of an option is an integer in $\{1, \dots, P\}$, P being the number of preference levels. 1 is the value associated with the customer’s preferred option and P corresponds to the least preferred option.
- The *service duration* of an option represents the time needed to deliver the customer’s package after accessing the location.

Let us consider a homogeneous fleet of vehicles starting from a given depot and returning to this depot within the time period. Travel time and routing costs between each pair of locations are assumed to be known.

The VRPDO consists in designing a route for each vehicle, serving each customer with one of his/her options, and satisfying the SDL capacities and minimal service level constraints. For each level $p \in \{1, \dots, P-1\}$, the service level SL_p of a solution is defined as the percentage of customers served with an option of level at most p. A minimal service level β_p has to be achieved for each level $p \in \{1, \dots, P-1\}$.

Similarly to many vehicle routing problems, the objective function is the minimization of the number of vehicles and the sum of routing costs, in lexicographical order. Note that each customer’s order has to be delivered by a single truck (i.e. split deliveries are not allowed).

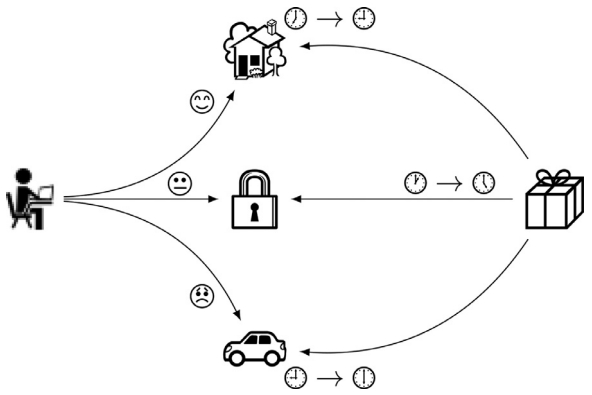


Fig. 1. Example of a customer's preferences in the VRPDO. The delivery options, in decreasing preference order are: at home, between 7 p.m. and 9 p.m., in a locker between 1 p.m. and 5 p.m., and in the car's trunk, between 9 a.m. and 6 p.m.

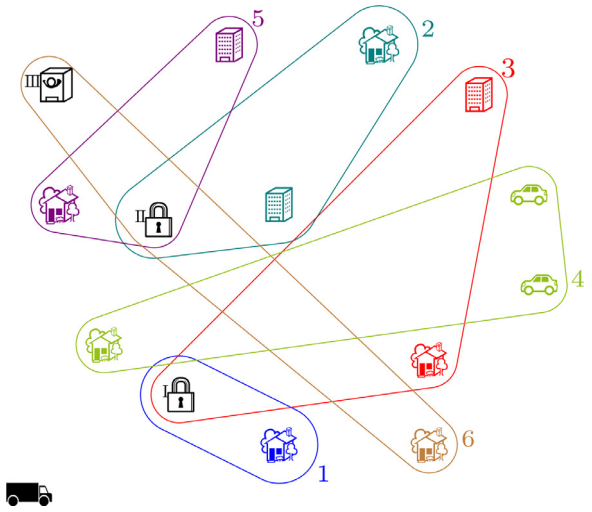


Fig. 2. An instance of the VRPDO from the carrier perspective. Individual delivery locations include home, office and car's trunk. Shared delivery locations include shops and lockers.

When a customer orders a parcel, he/she chooses multiple options, sorted by preference level. In the VRPDO, the time windows relate to the delivery of parcels by vehicles. At individual locations, these time windows are selected by customers. At SDLs, they correspond to an interval of time during which the corresponding facilities can be accessed by vehicles. Thus, it is assumed that all options delivered at SDLs have the same time windows. In practice, once a customer's order is delivered at an SDL or a remote location (i.e. car's trunk), the customer is notified ; he/she can collect the package at any time after being notified, including on the next days. Hence, once an order is delivered at an SDL, it is considered to occupy a part of the SDL capacity until the end of the period. There might be several policies to manage uncollected parcels, but this question is out of the scope of the present paper. Thus, we simply assume that the capacity of an SDL is *a priori* adjusted before solving the VRPDO, according to the parcels remaining at this location.

To illustrate the definition of the VRPDO, Fig. 1 shows the options from the customer perspective. Fig. 1 shows the example of a customer with 3 delivery options. Delivery at home between 7 p.m. and 9 p.m. is the preferred option, then delivery in a locker between 1 p.m. and 5 p.m., then a delivery in the trunk of his/her car between 9 a.m. and 6 p.m.

Fig. 2 presents the options from the carrier perspective. The individual options of each customer are represented by a group of icons of identical colors. SDLs are represented by black icons (locker, store/postal office). Each customer is represented by a number and its options are surrounded by a line of the same color as the icons. The carrier has to serve all customers via exactly one option. This amounts to define a set of routes that start from the carrier's depot (represented by the truck icon) and visit exactly one option inside a colored line.

Like in the Generalized Vehicle Routing Problem (GVRP) (Ghiani and Improta, 2000), the set of options is composed of subsets, for each customer. Vehicles must visit exactly one option in each subset. In the GVRP, subsets of options form a

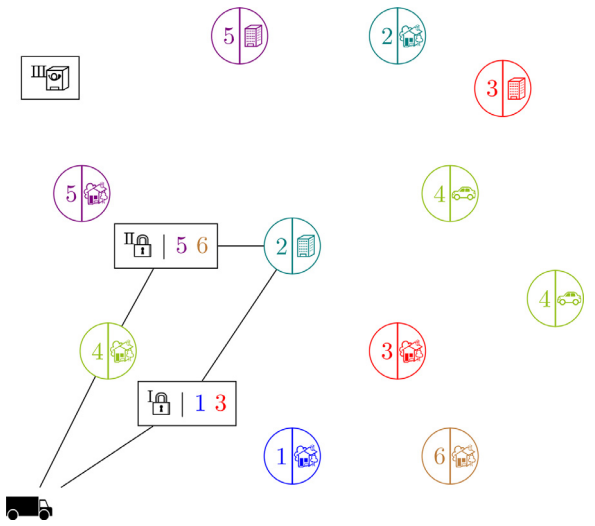


Fig. 3. Example of a route in the VRPDO. Customers 1 and 3 are now delivered at locker station I; customers 5 and 6 are delivered at locker station II.

partition of the set of locations. In the VRPDO, the subsets of options form a cover of the set of locations. This is because options associated with distinct customers can take place at the same shared location. An instance of the Generalized Vehicle Routing Problem with Time Windows (GVRPTW) (Moccia et al., 2012) is an instance of the VRPDO without shared locations and a single preference level. Hence, the VRPDO can be seen as a generalization of the GVRPTW.

Fig. 3 represents a solution to the VRPDO instance introduced in Fig. 2, with only one route. An option that takes place in an individual location is depicted by a circle including the customer's identity and the location. This is the case of customer 2, who is delivered at office rather than at home, thus avoiding a big detour. A shared location is depicted by a rectangle that includes the locations and identifies the delivered customers. For example, customers 1 and 3 are delivered at SDL I.

The time spent by a vehicle at a location is equal to the preparation time of the location plus the service duration of the visited options. The preparation time of a location corresponds to the time to find a parking place. Hence, it is counted only once, regardless of the number of parcels delivered. The service duration of an option corresponds to the time to deliver the parcel or to put it in a box.

Fig. 4 represents a solution to the VRPTW for the same instance, using home delivery only.

We assume that a parcel always occupies one box in a locker, no matter its size. Indeed, a locker is a rack of automatic boxes so there is at most one parcel per box, as long as it fits. Hence, the capacity of the shared locations, generally expressed as a number of boxes, can equivalently be expressed as a number of parcels. Due to uncertainty on when customers will pick up their parcels, it is assumed that each box can be used only once per day. On the contrary, it is realistic to consider that some shared locations (e.g. post offices) have unbinding capacity.

The assignment of parcels to boxes is not taken into account in the VRPDO, but it would be possible to consider different sizes of boxes by duplicating options.

Fig. 5 illustrates the capacity constraints induced by shared locations for the example described in Fig. 3. Not all shared locations must be visited. On the contrary, they can be visited by different vehicles.

Considering the quality of the service, Fig. 6 represents an example with a service level constraint requiring that at least $\beta_1 = 50\%$ of the customers are served with their preferred options and that at least $\beta_2 = 75\%$ of the customers are served with options of level 1 or 2. In the example, 50% of the customers are served with their preferred option (SL_1) and 83% of the customers are served with an option of level 1 or 2 (SL_2), so that the service level constraints are satisfied.

The constraints related to SDL capacity and service level constraints can be seen as resource constraints involving all the vehicles. They fall under the category of "synchronized resources" defined by Drexl (2012) as: "At any point in time, the total utilization or consumption of a specified resource by all vehicles must be less than or equal to a specified limit".

2.2. Mathematical model

Let N be the set of customers and O the set of options. The options of customer $c \in N$, denoted by $O_c \subset O$, are specific to that customer only. Thus, $\bigcup_{c \in N} O_c = O$ and $\forall c, c' \in N : c \neq c' \Rightarrow O_c \cap O_{c'} = \emptyset$.

Let L denote the set of locations and $O_l \subset O$ be the subset of options that take place at location $l \in L$.

The VRPDO is modeled on an option-based complete graph $G = (V, A)$. The set of vertices, denoted by $V = O \cup \{0, 0'\}$, contains one vertex for each option plus the starting depot 0 and the ending depot $0'$. The routes are defined on the option-

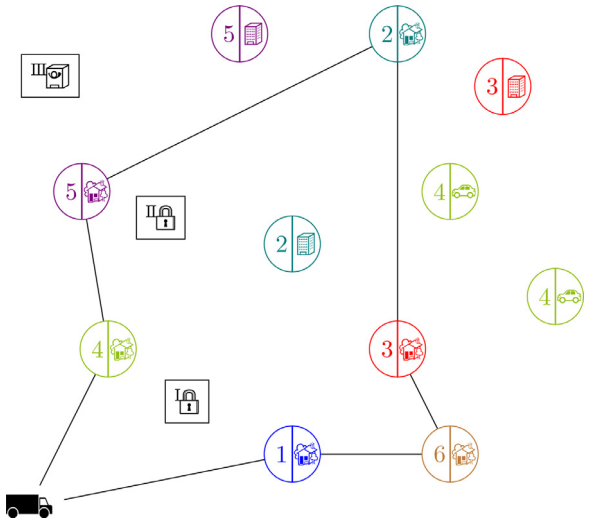


Fig. 4. Example of a route in the VRPTW, with home delivery only.

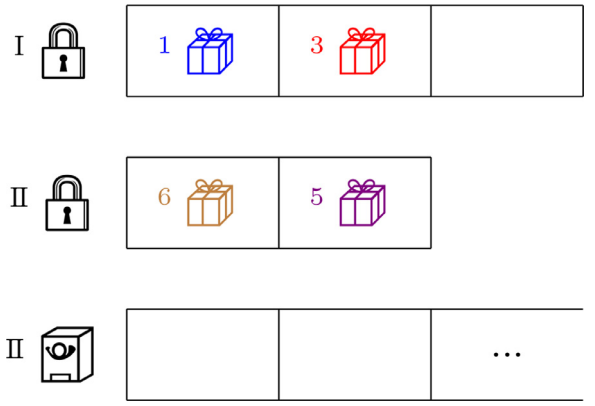


Fig. 5. Capacity at shared delivery locations (SDL). The locker stations I and II have limited capacity, while the capacity at stores is assumed unbounded.

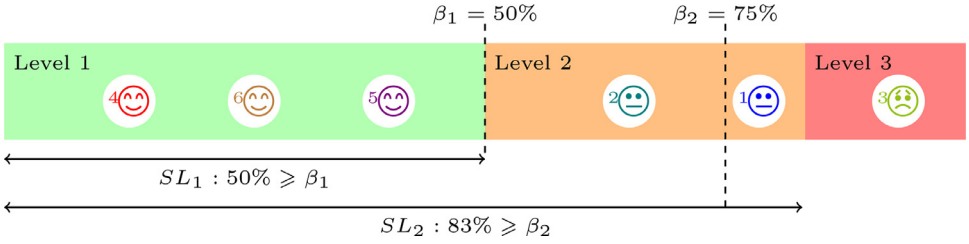


Fig. 6. Service level measured with respect to a set of selected options.



Fig. 7. Representation of the route from Fig. 3 on Graph G.

based graph G. A route is a sequence of options belonging to distinct customers. Consequently, when multiple deliveries are made in a shared location, it is represented by several vertices of V. Fig. 7 is the representation of the route from of Fig. 3.

The travel time on arc $(i, j) \in A$ is denoted by $t_{i,j}$. It includes the preparation time at the location of option j if the two locations are distinct. The cost of traveling on arc (i, j) is denoted by $c_{i,j}$. The service duration s_i at vertex $i \in V$ represents the time necessary to visit the option associated with this vertex. It is assumed to be null at depot vertices 0 and 0′. Each

option $i \in O$ is associated with the time window $[a_i, b_i]$ of its location. Moreover, it is assumed that $a_0 = a_{0'} = 0$ and that b_0 and $b_{0'}$ correspond to the end of the working day.

Let P be the number of preference levels. For a preference level $p \in \{1, \dots, P\}$, β_p is the minimal percentage of customers that can be served with an option of level p or lower. By definition $\beta_P = 100\%$ and $\forall p \in \{1, \dots, P-1\} : \beta_p \leqslant \beta_{p+1}$.

Let C_l be the capacity of location $l \in L$, expressed as a number of parcels. Each option $o \in O$ is associated with a preference level p_o and a demand q_o which is equal to the demand of its associated customer. The set of vehicle is denoted by K. It is assumed homogeneous with vehicles' capacity Q.

Model 1 describes the VRPDO. $x^k_{i,j}$ is a binary variable that indicates if the vehicle $k \in K$ uses arc $(i, j) \in A$. y_o is a binary variable that states whether option $o \in O$ is visited. h^k_i is the service time of vehicle $k \in K$ at vertex $i \in V$.

The first objective function (1) minimizes the number of vehicles. The second objective function (2) minimizes the routing costs. Constraints (3) state that exactly one option must be selected for each customer. Constraints (4) represent the satisfaction of vehicle capacity. Recalling that the capacity of SDLs is expressed as a number of parcels, independently of their size, constraints (5) state that the number of options delivered at location $l \in L$ cannot exceed its capacity C_l. Constraints (6) model the satisfaction of preference levels. For each preference level $p \in \{1, \dots, P\}$, at least $\beta_p\%$ of the total number of customers must be served with an option of level p or better. Constraints (7) state that an option can be used only if it is visited by one vehicle. The remaining constraints are the classical VRPTW constraints (Cordeau et al., 2002): Constraints (8) ensure that a vehicle that leaves the depot returns it at the end of its route, Constraints (9) ensure the continuity of the routes, Constraints (10) compute the service time and Constraints (11) ensure the respect of the time windows. M is an arbitrarily large positive value.

Model 1: option-based graph model

$$lex - min(z_1, z_2)$$

$$s.t.\ z_1 = \sum_{j \in V} x^k_{0,j} \tag{1}$$

$$z_2 = \sum_{k \in \{1,\dots,K\}} \sum_{(i,j) \in A} c_{i,j} x^k_{i,j} \tag{2}$$

$$\sum_{o \in O_c} y_o = 1 \qquad \forall c \in N \tag{3}$$

$$\sum_{o \in O} \sum_{(i,o) \in A} x^k_{i,o} q_o \leqslant Q \qquad \forall k \in K \tag{4}$$

$$\sum_{o \in O_l} y_o \leqslant C_l \qquad \forall l \in L \tag{5}$$

$$\sum_{o \in O | p_o \leqslant p} y_o \geqslant \beta_p \times |N| \qquad \forall p \in \{1, \dots, P-1\} \tag{6}$$

$$\sum_{(i,o) \in A} \sum_{k \in K} x^k_{i,o} \geqslant y_o \qquad \forall o \in O \tag{7}$$

$$\sum_{j \in V} x^k_{0,j} = \sum_{i \in V} x^k_{i,0'} \qquad \forall k \in K \tag{8}$$

$$\sum_{(i,j) \in A} x^k_{i,j} - \sum_{(j,i) \in A} x^k_{j,i} = 0 \qquad \forall k \in K, \forall j \in O \tag{9}$$

$$h^k_i - h^k_j + t_{i,j} + s_i \leqslant M.(1 - x^k_{i,j}) \qquad \forall k \in K, \forall (i,j) \in A \tag{10}$$

$$a_i \leqslant h^k_i \leqslant b_i \qquad \forall k \in K, \forall i \in V \tag{11}$$

$$x^k_{i,j} \in \{0, 1\} \qquad \forall k \in K, \forall (i,j) \in A$$

$$y_o \in \{0, 1\} \qquad \forall o \in O$$

$$h^k_i \geqslant 0 \qquad \forall k \in K, \forall i \in V$$

3. Relevant literature

The VRPDO is a VRPTW with multiple delivery options and synchronized resources. This section is organized according to these three aspects: VRPTW, delivery options and resource synchronization. In addition, some closely related problems are laid out.

3.1. VRPTW

Since the introduction of the VRPTW by Savelsbergh (1985), a plethora of papers have been published on this subject. We refer to Bräysy and Gendreau (2005a,b) for a review of the applicable literature and to Vidal et al. (2013) for the latest advances. Many LNS heuristics have been used to solve problems related to the VRPTW. They will be briefly discussed in Section 4.

3.2. Delivery options

In the VRPDO, customer requests can be satisfied at various locations. This characteristic is shared by the Generalized Vehicle Routing Problem with Time Windows (GVRPTW) (Moccia et al., 2012). The GVRPTW is a specific case of the VRPDO without shared locations and with a single preference level. The literature on the GVRPTW is very scarce. We refer to Afsar et al. (2014) for details on the GVRP, and to Moccia et al. (2012) for details of the GVRPTW. Yuan et al. (2018) developed a branch-and-cut for the GTSPTW (a single vehicle GVRPTW) that can solve instances with 30 clusters. Moccia et al. (2012) proposed a tabu search able to tackle instances with 120 clusters within few minutes.

The Vehicle Routing Problem with Multiple Time Windows (VRPMTW) (Favaretto et al., 2007) can be seen as a special case of the VRPDO and of the GVRPTW, where all options of a customer take place at the same location and their time windows are disjointed. For more detail, see Tricoire et al. (2010), Belhaiza et al. (2014) and Hoogeboom and Dullaert (2019).

In addition to the GVRP, some classes of vehicle routing problems include choices in the locations to visit. The goal of these problems is to select locations to visit and find routes between these locations, such that requirements are met at minimum cost or such that profit is maximized. This category of problem includes the Team Orienteering Problem (TOP), the VRP with profit (Archetti et al., 2014; Vansteenwegen et al., 2011), the Traveling Purchaser Problem (TPP) (Bernardino and Paias, 2018), and the Covering Tour Problem (CTP) (Kammoun et al., 2017).

Among the heuristics that have been proposed to solve VRP with choices in locations or time windows, most of the authors include a perturbation component in their method. For variants of the TOP, Souffriau et al. (2013) combined the Greedy Randomized Adaptive Search Procedure (GRASP) with an Iterated Local Search (ILS) and Tricoire et al. (2010) used a Variable neighborhood Search (VNS) metaheuristic. For variants of the CTP, Allahyari et al. (2015) combined GRASP with ILS, Takada et al. (2015) used ILS, Vargas et al. (2015) worked with an Adaptive Large neighborhood Search (ALNS) and Kammoun et al. (2017) proposed a VNS.

Besides, some papers propose using dynamic programming. In the context of a multi-period TOP with multiple time windows, Tricoire et al. (2010) used dynamic programming to choose which time window should be used in a given route. Moccia et al. (2012) solved a GVRPTW; when a customer is inserted in a route, the options for the other customers of this route can be changed through dynamic programming. Vargas et al. (2015) solved the CTP with an ALNS algorithm that uses dynamic programming to decompose a giant tour into routes.

3.3. Resource synchronization

Drexl (2012) underlined that Hempsch and Irnich (2008) was almost the only paper considering synchronized resources. Few other heuristics have been proposed since then. In Hempsch and Irnich (2008), Grangier et al. (2017b), Froger et al. (2017), the resources are only temporarily used during a limited period of time, before becoming available again. For example, in Grangier et al. (2017b), a truck uses a dock at a satellite facility only during the loading and unloading operations.

In the VRPDO, the synchronized resources are permanently used during the whole time horizon. Thus, it is only necessary to count the number of parcels in each locker and the number of visited options of each preference level to check the satisfaction of the synchronized resource constraints.

Souffriau et al. (2013) defined a variant of the TOP applied to tour planning for tourists. It is called the Multiconstraint Team Orienteering Problem with Multiple Time Windows (MC-TOP-MTW). The routes represent the different days of the trip. The synchronized resource constraints represent the budget and the maximal number of monuments of each type that can be visited. These resources are considered in a local search with a label on the locations. These labels are updated at each modification of the schedule. To get good solutions even with these resources, their algorithm relies on perturbations through a combination of GRASP with ILS. Besides, local search moves are chosen with a score. It is a function of the increase in the objective function, time shift, and resource consumption.

3.4. Closely related problems

Now that the three components of the VRPDO have been described, let us focus on very close problems.

Reyes et al. (2017) defined the VRP with Roaming Delivery Locations (VRPRDL). This problem involves delivering parcels into the trunks of cars, which move according to known schedules. In the VRPRDL, there are no synchronized resources, the time windows of a customer are disjointed and cars do not move faster than the delivery truck. This problem is solved with an LNS heuristic. Classical operators are adapted and dynamic programming is used to re-optimize routes by selecting better options without changing the customer sequence in the route. The combination with home delivery is considered

in a variant called VRP with Home and Roaming Delivery Locations (VRPHRDL). It is defined by Ozbaygin et al. (2017) and solved to optimality by a Branch-and-Price (B&P) algorithm for instances with up to 60 customers. Note that the solution approach in Ozbaygin et al. (2017) can also be used to solve instances of the GVRPTW.

Sitek and Wikarek (2017) defined the Capacitated VRP with Pick-up and Alternative Delivery (CVRPPAD). This paper considers multiple options for each parcel to be delivered. Lockers and post offices are modeled with limited capacities. Customer preferences are modeled with a penalty on the objective function if a non-desired option is used. No time windows are considered. Their method relies on a pre-processing phase done by constraint programming. A heuristic then groups parcels together before assigning them to a route.

Zhou et al. (2018) defined the Multi-Depot Two-Echelon Vehicle Routing Problem with Delivery Options (MD-2EVRP-DO). In this paper, a parcel can either be delivered at the customer's home or to a selected pick-up facility. Shared locations are incapacitated and customer preferences are modeled through penalties in the objective function. Time windows are not considered. This problem is solved with a multi-population genetic algorithm that embeds an ad hoc local search.

The most closely related study seems to be the one by Yuan et al. (2019), on the GVRPTW. Their metaheuristic method is based on a Set Partitioning formulation of the problem, coupled with an LNS heuristic and a local search that works with dual information.

4. Solution method

The Large Neighborhood Search (LNS) metaheuristic was first proposed by Shaw (1998) in a constraint programming context. In LNS, the current solution is iteratively improved by ruining it (i.e. removing a part of it) and recreating it (i.e reinserting the removed parts). This process is repeated until a stopping criterion (usually a time limit). The potential of LNS for solving a large variety of vehicle routing problems was revealed by Ropke and Pisinger who proposed an adaptive LNS, consisting of multiple search operators adaptively selected according to their past performance (Ropke and Pisinger, 2006b; 2006a; Pisinger and Ropke, 2007). LNS has been successfully applied to many variants of vehicle routing problems (Pisinger and Ropke, 2019) and the literature is abundant. Recently, Turkeš et al. (2019) compiled many papers using an LNS heuristic in a meta-analysis that shows the limited impact of the adaptive component proposed by Ropke and Pisinger (2006b).

We have selected a few papers of major importance with respect to the VRPDO: Shaw (1998) (VRP, VRPTW), Ropke and Pisinger (2006b,a); Pisinger and Ropke (2007) (diverse VRPTW), Nagata and Bräysy (2009) (VRPTW), Prescott-Gagnon et al. (2009) (VRPTW), Demir et al. (2012) (VRPTW and variants) and Christiaens and Vanden Berghe (2020) (CVRP and variants). Almost all operators from these papers were implemented, so that we have a representative panel of the LNS operators. In addition, ad hoc operators for the VRPDO were developed in order to integrate synchronized resources and shared locations. The 20 ruin operators implemented are presented in Section 4.2 and the 15 recreate operators implemented are presented in Section 4.3. Numerous tests were performed, using a rigorous methodology, to restrict the operators used in our LNS implementation. The experimental procedure to selected operators is presented in Section 4.4. For the sake of readability, in Sections 4.2 and 4.3, the operators that were selected are listed with an [S] and the ones that were discarded are listed with a [D], with respect to the results explained in Section 4.4. Note that discarded operators may be as efficient as selected ones, but redundant in our configuration. For the sake of traceability, the original names have been used whenever possible.

Sections 4.5 and 4.6 present the acceptance criterion and the set partitioning problem, respectively. The final tuning of the presented algorithm is detailed in Section 4.7.

4.1. Large neighborhood search heuristic

Algorithm 1 presents the main steps of the proposed heuristic. It is an LNS metaheuristic wherein a Set Partitioning Problem (SPP) is periodically solved. We, therefore, call it LNS-SPP. The main loop of the iterative process is from lines (3) to (18). In line (5), a ruin operator σ^- is randomly selected from a set Σ^- of operators. Each operator has a given constant probability of being selected. The same process is used in line (6) to select a recreate operator σ^+ from a set Σ^+ of operators. In line (7), the size Φ of the destruction is randomly chosen in a given interval $[\delta, \Delta]$. The destruction size is the percentage of customers to be removed from the current solution.

The selected operators are applied to the current solution s' in line (8). First, Φ% of the customers are removed from the solution with the chosen ruin operator σ^-. These customers are placed in the so-called request bank. Second, the customers from the request bank are inserted in the solution by the recreate operator σ^+. In line (9), the routes of the newly generated solution are stored into a set of routes: the pool of routes R. In line (10), an acceptance criterion is used to decide whether the new solution becomes the current solution for the next LNS iteration. The routes of pool R are recombined every η iterations by solving an SPP in line (14). After this combination, the pool of routes R is emptied and re-initialized with the routes of the best solution, in line (15).

The VRPDO has two lexicographic objectives: (1) to minimize the number of vehicles, and (2) to minimize the routing costs (described by the objective function 2). Similarly to Ropke and Pisinger (2006a), Algorithm 1 is run twice, with half of the time budget for each part. During the first phase, the number of vehicles is decreased by removing the smallest route from the solution each time a feasible solution is found (the customers are placed in the request bank). During the second

Algorithm 1: LNS-SPP.

Parameters: a set Σ^+ of recreate operators, a set Σ^- of ruin operators, a frequency η
Input: an initial solution s

pool of routes $R = \emptyset$
nbIt $= 1$
while the time budget is not reached **do**
  $s' \leftarrow s$
  randomly select a ruin operator $\sigma^- \in \Sigma^-$
  randomly select a recreate operator $\sigma^+ \in \Sigma^+$
  randomly select a destruction size $\Phi \in [\delta, \Delta]$
  $s' \leftarrow \sigma^+(\sigma^-(s', \Phi))$
  store the routes of s' into pool R
  if s' meets the acceptance criterion **then**
    $s \leftarrow s'$
  end if
  if nbIt$\%\eta = 0$ **then**
    $s =$ best combination of the routes of R found with a Set Partitioning Problem
    Empty R and re-initialize it with the routes of s
  end if
  nbIt $=$ nbIt $+ 1$
end while
return the best feasible solution found

phase, the routing costs are minimized, using the minimum number of vehicles found in a feasible solution obtained during the first phase. The configuration of our algorithm does not change between the two phases.

4.2. Ruin operators

Ruin operators use different rules to remove customers from the solutions. Upon removal, customers are placed in the request bank of the solution.

Ruin operators from the literature can be divided into two categories, named local ruin operators and large ruin operators. A local ruin operator deletes customers so that even if few customers are deleted, it is likely that the solution will be improved on reconstruction. On the contrary, a large ruin operator requires that more customers be deleted. Indeed, with a large ruin operator, if too few customers are removed, it is likely that they will be re-inserted in the same position.

4.2.1. Local ruin operators

The local ruin operators implemented from the literature are:

[S] *Distance-related removal* (Ropke and Pisinger, 2006b) : removes customers that are close to each other with respect to the Euclidean distance.
[D] *Node neighborhood removal* (Demir et al., 2012): removes customers that are close to each other with respect to the infinity norm.
[D] *Proximity removal* (Prescott-Gagnon et al., 2009): removes customers that are close both from a spatial and a temporal point of view, according to a parameterless formula. This is an extension of Shaw removal (Shaw, 1998).
[S] *(Split) String removal* (Christiaens and Vanden Berghe, 2020): removes sequences of customers in the routes of the current solution, either conserving, or not, a sub-string in the middle.

4.2.2. Large ruin operators

The large ruin operators implemented from the literature are:

[S] *Random removal* (Ropke and Pisinger, 2006b): randomly removes customers.
[D] *Demand-related removal* (Demir et al., 2012): removes customers that have demands of similar size.
[S] *Time-related removal* (Pisinger and Ropke, 2007): removes customers that are delivered at approximately the same time.
[S] *Zone removal* (Demir et al., 2012): randomly removes customers into predefined fixed rectangular zones.
[S] *Cluster removal* (Pisinger and Ropke, 2007): removes customers that are served by the same route in the current solution. A route is randomly selected and the Kruskal's algorithm is run on the arcs of this route until two clusters remain. All the customers in one of them, randomly chosen, are removed.
[S] *Route removal* (Nagata and Bräysy, 2009): removes all the customers of a route.
[D] *Distance worst removal* (Ropke and Pisinger, 2006a): iteratively removes the customer with the highest individual service cost. The individual service cost of a customer is the cost of the arcs that enter and exit the location where the given

customer is served in the current solution, minus the cost of going directly from the previous location on the route to the next one. If it is a shared location, this cost is divided by the number of customers served at this location on the same route.

[D] *Time worst removal* (Demir et al., 2012): removes the customers that cause the largest time loss.

[D] *Neighborhood removal* (Demir et al., 2012): first, the cost of each route divided by the number of customers served by this route is computed. The customers are then removed sequentially by decreasing order of the difference between their individual service cost and the average service cost of their route.

[D] *Node-pair history removal* (Pisinger and Ropke, 2007): memorizes the cost of the best solution that uses each arc. The operator removes the nodes that are reached via arcs with the largest score.

[S] *Historical knowledge node removal* (Demir et al., 2012): this history removal memorizes the lowest individual service cost of each customer. The operator removes the customers with the largest difference between their current individual service cost and their lowest individual service cost. It can be see as a history-biased worst removal.

Ruin operators for the VRPTW can be adapted to the VRPDO in two ways: option-based or customer-based. An option-based operator only takes in account the options that are currently visited to serve the customers. A customer-based operator takes all the options into account. Let us describe an example with *distance-related removal*. The distance between two customers in *distance option-based related removal* is the distance between the options that are currently visited to serve these customers. On the contrary, in *distance customer-based related removal*, it is the minimal distance between any two options of these customers. That is to say, the option-based version will remove customers that are currently served in close locations and the customer-based one will delete customers that are potentially served in close locations.

4.2.3. VRPDO specific ruin operators

The new ruin operators specifically developed for the VRPDO are:

[D] *Preference-oriented random removal*: randomly selects customers and deletes them with a probability based on the preference level of the options currently visited to serve them. The probability of deleting a selected customer, currently served with option o, is $(1/P + 1 - p_o)^3$. When there are three preference levels, the probability of deleting a customer served with an option of level 3, 2 and 1 is 1.0, 0.125 and 0.04, respectively.

[S] *SDL-oriented random removal* (Shared Delivery Location-oriented random removal): randomly selects customers in the solution. If the selected customer is delivered at an individual location, the probability of being deleted is only 10%. Otherwise, if customers are delivered at a shared location, they are always deleted.

[D] *Random SDL removal*: randomly selects a shared delivery location and removes all the customers delivered at this location.

[D] *SDL-related removal*: selects a shared delivery location and randomly deletes customers that have an option at this location.

[D] *SDL-worst removal*: a *distance worst removal* where the detour cost is fully assigned to all the customers served at the shared delivery locations. The detour cost is not divided by the number of customers served at this location.

All these ad-hoc ruin operators are large ruin operators. With the exception of the *SDL related removal*, they are all option-based.

4.3. Recreate operators

Most recreate operators follow the best insertion principle: any given customer is inserted at the position that minimizes the increase of routing costs. To compute the best insertion of customer c in route r, we try to insert all customer options in all positions of route r. Only feasible insertions are performed by the algorithm. Hence, a solution always satisfies the vehicle capacity constraints, time windows, shared location capacities, and service level constraints. The only form of infeasibility considered in LNS-SPP is the fact that not all the customers are served, i.e the request bank can be non-empty.

The forward time slacks (Savelsbergh, 1992) of all routes are stored in order to evaluate insertions in constant time with respect to time windows. The usage of each shared location and of each preference level is stored. Hence, testing the validity of insertions with respect to synchronized resources is done in constant time. When an insertion is performed, the forward time slacks of the corresponding route are updated in linear time with respect to the length of the route. The update of capacity usage is done in constant time.

4.3.1. Recreate operators from the literature

The LNS recreate operators from the literature can be divided into two categories: list heuristics and others. Most of the operators proposed in the literature may evaluate the insertion of a given customer into a given route multiple times. Typically, the insertion of each customer in each route is evaluated once at the beginning. After each modification, the insertion of the remaining customers in the modified route is then re-evaluated. In a list heuristic, the insertion of a customer into a route is evaluated at most once. Typically, the customers in the request bank are sorted once and inserted in this order. Consequently, list heuristics are very fast recreate operators.

The list heuristics implemented from the literature are:

[S] *Random order best insertions* (Christiaens and Vanden Berghe, 2020): sequentially inserts the customers in the request bank at their best insertion position in a random order.

[D] *Oldest first best insertions* (Christiaens and Vanden Berghe, 2020): sequentially inserts the customers in the request bank at their best insertion position in non-increasing order of the number of iterations since the last time a given customer was served.

[S] *Largest first best insertions* (Christiaens and Vanden Berghe, 2020): sequentially inserts the customers in the request bank at their best insertion position in non-increasing order of their demand.

[D] *Farthest first best insertions* (Christiaens and Vanden Berghe, 2020): sequentially inserts the customers in the request bank at their best insertion position in non-increasing order of their distance to the depot.

[D] *Closest first best insertions* (Christiaens and Vanden Berghe, 2020): sequentially inserts the customers in the request bank at their best insertion position in increasing order of their distance to the depot.

The other operators implemented from the literature are:

[D] *Best temporal insertions* (Demir et al., 2012): inserts the customers so that the loss of time is minimal. The loss of time is defined as the waiting time at the inserted option plus the waiting time at the next option on the route. Customers can be processed either in random order or by decreasing order of demand, respectively.

[D] *Greedy best insertion* (Ropke and Pisinger, 2006a): iteratively computes the cheapest insertion for each customer and inserts the customers that have the lowest insertion cost.

[S] *k-regret* (Ropke and Pisinger, 2006a): iteratively computes the best insertion cost on each route for each customer and inserts the one that has the largest difference between its best insertion cost and next $(k-1)$ route's best insertion costs.

[S] *Ejection search* (Nagata and Bräysy, 2009): first, all the customers in the request bank are placed in a FIFO structure. The customers from this structure are inserted into the solution by allowing some customers from the solution to be removed and put in the queue. As in Curtois et al. (2018) the procedure is heuristically sped up. First, to insert one customer, at most two customers can be removed from the solution. Second, insertions are tested with an increasing number of removed customers. If a feasible insertion is found, insertions with more removals will not be tested. Third, when customers must be removed to insert the customer in question, the insertion that removes the customers with the smallest score is chosen. The score of a customer is the number of times where no feasible insertion was found for this customer during all the calls to *ejection search*. Finally, the number of iterations of *ejection search*, at each call, is limited to five times the initial size of the request bank.

4.3.2. VRPDO specific recreate operators

The operators specifically developed for the VRPDO are:

[S] *Preferred best insertion*: this operator is an adaption of the classical best insertion operator. The only difference is that the insertion costs are modified in a lexicographical manner: the cost of all insertion positions with all delivery options are calculated, then these insertions costs are lexicographically sorted by considering the preference levels p_o first and the insertion costs second.

[D] *Preference regret*: the best insertion is computed for each customer and for each preference level. Let C_p^i be the cost of the best insertion of a customer i with an option of level p or lower. The *preference regret* score of customer i is $\sum_{p=1}^{P-1}(C_P^i - C_p^i)$. Customers are always inserted at their cheapest position and the customer with the largest regret is inserted first.

[D] *Normalized best insertion*: the equalized insertion cost is the real cost of insertion, divided by the capacity of the location. *Normalized best insertion* is a *greedy best insertion* that uses an equalized insertion cost to select the insertion possibility for each customer and select the first customer to insert.

[S] *SDL-regret* (Shared Delivery Location regret): customers are inserted at their cheapest insertion by decreasing order of their regret. In this version, the regret of a customer is the difference between the insertion cost when all options are allowed or the cost when only individual locations are authorized. For example, let us consider the partial solution represented in Fig. 8 (same instance as in Fig. 3). Customers 5 and 6 are not served. Both can be delivered at locker II, like customer 4. But this locker only has a capacity of two, as in Fig. 5. For both customers 5 and 6, the cheapest insertion is in this locker, with a cost of 0. Figs. 9 and 10 show the cheapest insertion of customers 5 and 6 without considering shared locations. Hence, SDL-regret for customer 6 is higher than that of customer 5. In this case, the *SDL-regret* will select customer 6 first and insert him/her in the locker.

4.4. Operator selection

As described in sections 4.2 and 4.3, a number of operators have been implemented. Because it does not seem useful to keep them all, we searched for a configuration of LNS-SPP with fewer operators. In this section, we define a *configuration* as a subset of ruin operators and a subset of recreate operators.

A statistical study was performed to choose a configuration from the 20 ruin operators and the 15 recreate operators that were implemented. Tests were performed on a representative set of randomly generated VRPDO instances. To determine

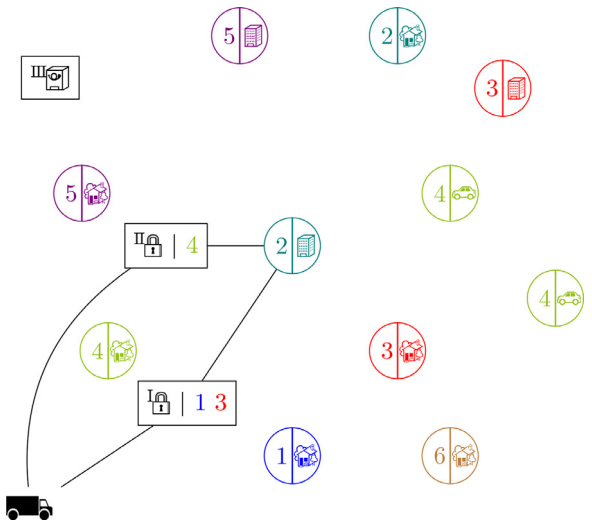


Fig. 8. Example of a partial solution.

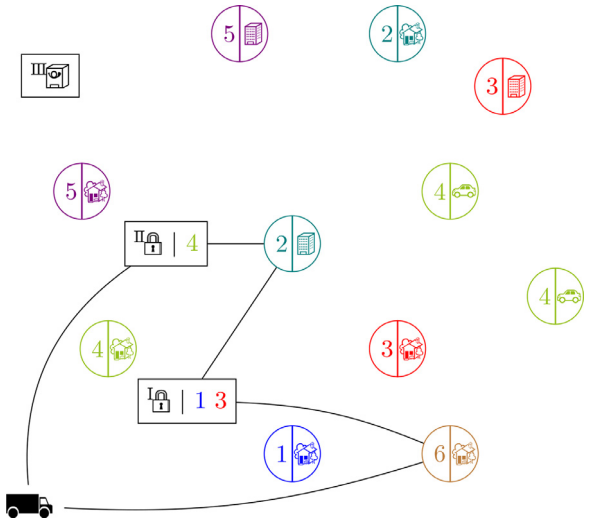


Fig. 9. Insertion of the brown (6) customer without lockers.

whether one configuration is significantly different from another, the Wilcoxon pairwise test (Wilcoxon, 1945) was used with a threshold of 5%. This test compares two populations of results. In our case, it compares the results obtained by two configurations on each instance of a set of VRPDO instances. This methodology is inspired from Stützle (2018). Note that the automatic configuration package IRACE (López-Ibánez et al., 2011) was also used, but it did not converge after several days of computation.

Let us define a class of operators as a group of operators that have similar purposes. First, the ruin operators are split into 4 classes: random removals, related removals, worst removals and history removals. Second, the recreate operators are split into 4 classes: list heuristics, time best insertion heuristics, regret heuristics and ejection search. This classification is detailed in Tables 1 and 2.

This statistical study is decomposed in two phases: (1) a study of the impact of each class of operators; (2) a study of the impact of each individual operator.

To study classes of operators, the reference configurations are: the full configuration with all the operators, and a minimal configuration with as few operators as possible. All the operators of each class are removed from the full configuration and added to the minimal configuration. All these “sub-configurations” were tested on all instances of the test set and compared. This first phase determines whether the considered operators are redundant with other operators of the full configuration, and whether they improve the results of the minimal configuration.

Based on the previous results, we build an intermediate configuration. The operators from this configuration are changed one by one. If a given operator was used, then it is deactivated, otherwise it is added to the configuration. All these alter-

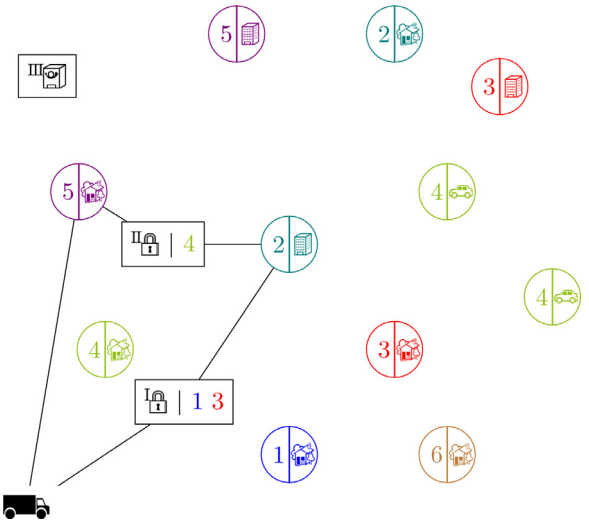


Fig. 10. Insertion of the purple (5) customer without lockers. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Overview of configuration experiments for ruin operators.

	Option-based operators	Customer-based operators
Random removals	- preference-oriented random removal + SDL-oriented random removal - random SDL removal	+ random option removal
Related removals	- distance-related option removal - node neighborhood removal - time-related option removal + (split) string removal + cluster removal + route removal	+ distance-related customer removal + zone removal + time-related customer removal - proximity customer removal - demand-related removal - SDL-related removal
Worst removals	- distance worst removal - time worst removal - neighborhood removal - SDL worst removal	
History removals	- node-pair history removal	+ historical knowledge node removal

Table 2
Overview of configuration experiments for recreate operators.

	VRPTW operators	VRPDO operators
List heuristics	+ random order best insertion - oldest first best insertion + largest first best insertion - farthest first best insertion - closest first best insertion	- normalized best insertion + preferred best insertion
Time best insertions	- time best insertion	
Regrets	- greedy best insertion + 2-regret - 3-regret - 4-regret	+ SDL-regret - preference regret
Ejection search	+ ejection search	

native configurations are compared with the intermediate configuration. It determines if the assessed operator significantly improves the results, or if it is redundant with used operators.

The results of the operators are summarized in Tables 1 and 2. Operators marked with $^{++}$ are considered essential. Operators marked with $^{+}$ slightly improve the results. Operators marked with $^{-}$ are useless or redundant with already kept operators, and adding them does not improve the results. The selected operators are those rated $^{++}$ and $^{+}$.

4.5. Acceptance criterion

In line 10 of Algorithm 1, the acceptance criterion determines whether the newly generated solution should be accepted as the current solution at the next iteration.

Ropke and Pisinger (2006b) used the Metropolis criterion from Simulated Annealing (Kirkpatrick et al., 1983). To deal with partial solutions, a modified cost was proposed in Pisinger and Ropke (2007). The modified cost of Equation (12) penalizes the unserved customers with factor ζ. In this formula, B is the request bank of the current solution, *cost* is its routing cost (described by 2) and N is the set of customers.

$$\text{modified cost} \;=\; \text{cost} \times \left(1 + \zeta . \frac{|B|}{|N|}\right) \tag{12}$$

Santini et al. (2018) conducted a comprehensive study on the acceptance criteria for the LNS metaheuristic. In their conclusion about the CVRP, they advocated for the record-to-record criterion (Dueck, 1993). With this criterion, the solution is accepted if its modified cost is less than T% larger than the modified cost of the best known solution. Furthermore, they propose decreasing the acceptance threshold T during the algorithm. T decreases linearly between its initial value at the beginning and 0, when the time limit is reached. They conclude that the best values of T and ζ depend of the type and size of instance.

Our implementation uses the record-to-record criterion with modified cost (12). To avoid tuning parameters and get a reliable acceptance criterion, a simple adaptive procedure is proposed.

Our experiments empirically show that LNS-SPP performs well if the ratio of accepted solutions is between 4% and 14%. T and ζ are changed so as to maintain the ratio of accepted solutions in this target. The ratio of accepted solutions is periodically evaluated. If the ratio of accepted solutions is less than 4%, T is multiplied by 1.5 and ζ is divided by 1.5. On the contrary, if it is larger than 14%, T is divided by 1.5 and ζ is multiplied by 1.5.

4.6. Set partitioning problem

The utilization of the Set Partitioning Problem (SPP) to solve vehicle routing problems was first introduced by Foster and Ryan (1976). It is now widely used in column generation methods (Toth and Vigo, 2014) for many routing problems (Archetti et al., 2014). It can also be used to recombine routes that are produced by a heuristic. The SPP is used as a post-optimization technique (Rochat and Taillard, 1995; Mancini, 2017; Gschwind and Drexl, 2019), as well as inside hybrid heuristics (Prescott-Gagnon et al., 2009; Groër et al., 2011; Mendoza and Villegas, 2013; Subramanian et al., 2013; Parragh and Schmid, 2013; Yıldırım and Çatay, 2015; Grangier et al., 2017a; Tellez et al., 2018).

We solve the SPP by solving a Set Covering Problem (SCP) formulation and by repairing the solution if a customer is served more than once. Yıldırım and Çatay (2015) showed that solving an SCP instead of a SPP slightly shortens solving time. Additionally, an exact reparation is rarely needed and a greedy procedure finds the optimal reparation almost all the time.

The SCP model for the VRPDO is represented in Model 2. R is a set of routes that are valid with respect to time windows and vehicle capacity. Let us define R as the pool of routes generated in Algorithm 1. Indicator α_r^o has value 1 if the option $o \in O$ is visited by the route $r \in R$; it is equal to 0 otherwise. The cost of route $r \in R$ is denoted by w_r. Let z^r be a binary variable that indicates whether route $r \in R$ is used in the solution.

The objective function (13) minimizes the sum of the cost of the routes used. Constraints (14) state that each customer must be served at least once. Constraints (15) express the capacity of the shared locations. Constraints (16) are the service level constraints. Because it is a set covering formulation, more than one option may be used to serve a customer. Consequently, the service level constraint is expressed as a capacity constraint. Constraints (15) and (16) are the synchronized resource constraints. Constraints (17) set the upper bound on the number of vehicles used in the solution to $|K|$, which is the best known value for the minimum number of vehicles before solving the SCP.

Model 2: SCP for the VRPDO

$$min \sum_{r \in R} w_r z_r \tag{13}$$

$$s.c \sum_{r \in R} \sum_{o \in O_c} \alpha_r^o z_r \geqslant 1 \qquad \forall c \in N \tag{14}$$

$$\sum_{r \in R} \sum_{o \in O_l} \alpha_r^o z_r \leqslant C^l \qquad \forall l \in L \tag{15}$$

$$\sum_{r\in R}\sum_{o\in O|p_o\geqslant p}\alpha_r^o z_r \leqslant (1-\beta_{p+1})\times|N| \qquad \forall p\in\{1,\dots,P-1\} \tag{16}$$

$$\sum_{r\in R} z_r \leqslant |K| \tag{17}$$

$$z_r\in\{0,1\} \qquad \forall r\in R$$

Model 2 is solved every η iterations by an ILP solver. The set of routes R is then composed of the routes that have been generated by Algorithm 1 during the last η iterations. As proposed by Tellez et al. (2018), frequency η can be adapted. The value of η is reduced by a quarter when the solver does not succeed in proving optimality, nor in improving the best known solution, twice in a row. We extend this procedure as follows. η is increased by a quarter if the optimality is proven, or if the best-known solution is improved, twice in a row.

In addition, it is not necessary to solve the SPP every η iterations. While the LNS still improves the current solution, it is better to wait for stabilization. Hence, the SPP is solved only if LNS cannot improve the cost of the best-known solution by more than ρ% during the last η iterations.

We observe that the SCP defined for the VRPDO is more difficult to solve than the pure VRPTW set covering formulation. Thus, an extension of Model 2 is proposed to reduce solving time. Binary variables y_o are introduced to indicate whether option $o\in O$ is visited. These variables allow the solver to branch on options, hence discarding a lot of routes.

Constraints (18), (19), and (20) reformulate constraints (14), (15), and (16) with the y variables, respectively. Constraints (21) bind the two sets of variables by stating that if an option is visited, then the corresponding y_o must be set at value 1.

The following constraints are added to Model 2

$$\sum_{o\in O_c} y_o = 1 \qquad \forall c\in N \tag{18}$$

$$\sum_{o\in O_l} y_0 \leqslant C_l \qquad \forall l\in L \tag{19}$$

$$\sum_{o\in O|p_o\leqslant p} y_o \geqslant \beta_p\times|N| \qquad \forall p\in\{1,\dots,P\} \tag{20}$$

$$\sum_{r\in R}\alpha_r^o z_r \leqslant y_o \qquad \forall o\in O \tag{21}$$

$$y_o\in\{0,1\} \qquad \forall o\in O$$

4.7. Final parameter tuning

In our LNS-SPP, the probability of selecting each operator is constant throughout the algorithm. In the proposed implementation, all ruin operators are equiprobable. The probability of selecting each recreate operator is inversely proportional to its average running time. Furthermore, Christiaens and Vanden Berghe (2020) have recently proposed to perform a huge number of small and fast iterations in LNS in order to compensate the lack of local search in this metaheuristic. We added this functionality as a special case: if a list heuristic is selected, there is a high probability ϕ that the destruction size will be small (between δ_{mini} and Δ_{mini} percent of the customer) and that the ruin operator will be local, i.e *string removal, split string removal* or *distance-related customer removal*. The probability of each operator, δ, Δ, δ_{mini}, Δ_{mini} and ϕ was tuned according to the recommendations of IRACE (López-Ibánez et al., 2011).

The *blink* principle, introduced in Christiaens and Vanden Berghe (2020) for recreate operators, randomly ignores certain insertions with a given probability during the computation of the best insertion. In the proposed implementation, this feature did not prove to have a significant impact. It introduces diversification through randomization. Nevertheless, this principle has been applied to the ruin operators; for each removal evaluated there is a given probability of simply ignoring it.

To summarize, on the one hand, the list heuristics and the small destructions favor a high number of iterations. On the other hand, using regret heuristics and ejection search tends to reduce the number of iterations. As observed by Christiaens and Vanden Berghe (2020), the small destruction and list heuristic can compensate for a lack of local search. Furthermore, the numerous iterations coupled with blink provide a good exploration of the search space in the CVRP. To deal with time windows, it is worthwhile to perform larger destruction and to take some time to anticipate constraint violation, like Ropke and Pisinger (2006b) and Demir et al. (2012). Hence the final operator configuration in the proposed LNS-SPP combines fast and slow iterations in addition to long intensification phases with the SPP component.

To conclude this section, we indicate the values of the parameters (tuned with the help of IRACE) :

- The probability of selecting each ruin operator is the same for all operators. The probability to blink a deletion possibility is 30%.

- The probability of selecting a recreate operator is inversely proportional to its running time. The probability of selecting list heuristics (*random order best insertion, largest first best insertion*) is 0.4. The probability of selecting the other recreate operator (*2-regret, ejection search, SDL-regret* and *preferred best insertion*) is 0.05.
- The destruction size and removal operator selection rule is different for the list heuristics and the other recreate operators. In general the destruction size is between $\delta = 10\%$ and $\Delta = 20\%$ of the number of customers. For the list heuristic there is a 30% probability of performing a classical destruction (using any ruin operator) and a $\phi = 70\%$ probability of performing a small, local destruction. That is to say, only between $\delta_{\text{mini}} = 1\%$ and $\Delta_{\text{mini}} = 10\%$ of customers are removed, and a local removal operator (*distance-related removal, (split) string removal*) is used.
- The initial values of the record-to-record acceptance criteria are $T = 0.18$ and $\zeta = 9$. These values are adjusted every 4 500 iterations by a factor of 1.5, as described in Section 4.5.
- The initial call frequency to the SPP component is $\eta = 20000$. It is adjusted by a factor of 1.25. Furthermore, the time budget for the solver is 30 seconds and $\rho = 1\%$.

5. Experiments

The method is coded in C++ and is compiled with g++ 5.4.0. The MIP models are solved with IBM Ilog CPLEX 12.8.0, with the following options: branch up first and emphasis on hidden feasible solutions. The experiments were performed using Linux, Ubuntu 16.04 LTS, running on an Intel Xeon X5650 @ 2.57 GHz. A single core is used by our code and the third-party solvers.

To the best of our knowledge, no instances exist for the VRPTW with synchronized resource constraints. Section 5.1 validates the proposed matheuristic on related problems. Section 5.2 presents the generated VRPDO instances and Section 5.3 presents managerial insights.

5.1. Validation of LNS-SPP

The VRPDO instances described in Section 5.2 were recently generated by Olkis et al. (2020). The authors solved instances with up to 50 customers integer distances and small time-windows, with a branch-and-cut-and-price approach. For the 16 instances considered, the average gap reported in Olkis et al. (2020) between the lower bound and the upper bound is 1.15% after one hour of computation. The average gap of the LNS-SPP, out of five runs, is 0.30% after 30 seconds of computation. This shows the efficiency of the LNS-SPP on small instances of the VRPDO.

In order to assess the efficiency of the LNS-SPP algorithm on larger instances, we also used it to solve two closely related optimization problems: the VRP with time windows (VRPTW) and the VRP with roaming delivery location.

5.1.1. Results on the vehicle routing problem with time windows

The LNS-SPP was evaluated on 176 benchmark instances of the VRPTW, proposed by Solomon (Solomon and Desrosiers, 1988) (100 customers) and Gehring and Homberger (Gehring and Homberger, 1999) (200 and 400 customers). Instances of type R are composed of randomly located customers. Instances of type C are composed of clustered customers. Instances of type RC are a mix of random and clustered customers. Each type of instance includes two groups, in which routes visit a few customers and many customers, respectively.

Tables 3, 4 and 5 present the results for instances with 100, 200 and 400 customers, respectively. In each table, the results found by LNS-SPP are compared with those obtained by the ALNS of Pisinger and Ropke (2007) and by the state of the art Hybrid Genetic Algorithm with Adaptive Diversity Management (HGSADC) of Vidal et al. (2013). The time limits of each run of LNS-SPP is close to those used in Vidal et al. (2013). Each line represents the average result for one type of instance. Instance groups are listed in column 1. Columns 2, 3 represent the average number of routes in the solution of the ALNS and the average cost, over all instances of each group (only the best result out of 5 runs is considered). Columns 4, 5, and 8, 9 give the same information for the HGSADC and the LNS-SPP, respectively. Additionally, columns 6, 7 depict the

Table 3
Solomon VRPTW instances with 100 customers: average number of routes and routing costs obtained with the ALNS Pisinger and Ropke (2007), the HGSADC Vidal et al. (2013) and the LNS-SPP.

Instance	ALNS		HGSADC		LNS-SPP			
group	Best of 5		Best of 5		Average of 5		Best of 5	
C1	10.0	828.38	10.0	828.38	10.0	828.38	10.0	828.38
C2	3.0	589.86	3.0	589.86	3.0	616.81	3.0	589.86
R1	11.92	1 212.39	11.92	1 210.69	12.0	1 211.83	12.0	1 209.20
R2	2.73	957.72	2.73	951.51	2.75	965.73	2.73	956.00
RC1	11.5	1 385.78	11.5	1 384.17	11.53	1 384.83	11.5	1 384.17
RC2	3.25	1 123.49	3.25	1 119.24	3.25	1 141.36	3.25	1 123.17
Total	405	57 332	405	57 196	406.4	57 764	406	57 259
Time (*s*)	150		160		160			

Table 4
Homberger VRPTW instances with 200 customers: average number of routes and routing costs obtained with the ALNS Pisinger and Ropke (2007), the HGSADC Vidal et al. (2013) and the LNS-SPP.

Instance	ALNS		HGSADC		LNS-SPP			
group	Best of 5		Best of 5		Average of 5		Best of 5	
C1	18.9	2 718.77	18.9	2 718.41	18.9	2 726.79	18.9	2 720.12
C2	6.0	1 831.59	6.0	1 831.59	6.0	1 836.72	6.0	1 831.65
R1	18.2	3 615.69	18.2	3 613.16	18.2	3 638.43	18.2	3 625.09
R2	4.0	2 937.67	4.0	2 929.41	4.1	2 947.46	4.0	2 951.85
RC1	18.0	3 192.56	18.0	3 180.48	18.0	3 240.20	18.0	3 204.41
RC2	4.3	2 559.32	4.3	2 536.20	4.3	2 569.91	4.3	2 547.84
Total	694	168 556	694	168 092	694.6	169 595	694	168 809
Time (s)	3 180		504		600			

Table 5
Homberger VRPTW instances with 400 customers: average number of routes and routing costs obtained with the ALNS Pisinger and Ropke (2007), the HGSADC Vidal et al. (2013) and the LNS-SPP.

Instance	ALNS		HGSADC		LNS-SPP			
group	Best of 5		Best of 5		Average of 5		Best of 5	
C1	37.6	7 182.75	37.6	7 170.47	37.6	7 234.80	37.6	7 188.35
C2	11.9	3 874.58	11.6	3 952.95	11.9	3 900.55	11.9	3 877.18
R1	36.4	8 420.52	36.4	8 402.57	36.4	8 663.53	36.4	8 538.45
R2	8.0	6 213.48	8.0	6 152.92	8.0	6 262.88	8.0	6 238.56
RC1	36.0	7 940.65	36.0	7 907.14	36.1	8 253.79	36.0	8 145.91
RC2	8.6	5 269.09	8.5	5 215.21	8.7	5259.86	8.7	5 233.00
Total	1 385	389 011	1 381	388 013	1387.0	395 754	1386	392 214
Time (s)	5 340		2 046		3 000			

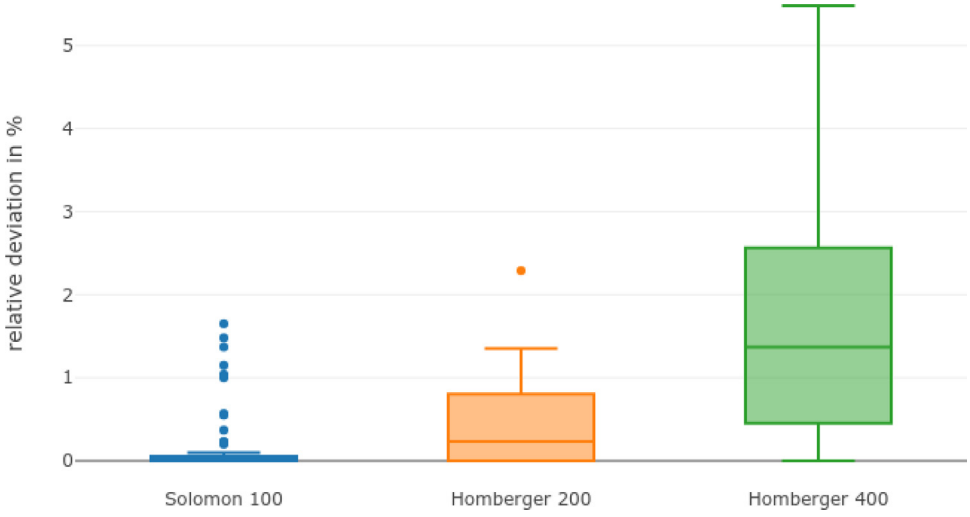


Fig. 11. Relative deviation between the results of the LNS-SPP and the best known solution, on various VRPTW instances.

average number of routes and routing costs, with LNS-SPP, averaged over the 5 runs on each instance. The last lines give the total over all instances for each algorithm considered, and the time budget allocated to each algorithm, respectively.

Fig. 11 shows the relative gap, in percentage, between the best known solutions and the best solutions found by the LNS-SPP out of 5 runs. The LNS-SPP algorithms finds the best known number of vehicles for all but six instances. These instances are not considered in this figure.

These experiments show that, although it has been tailored for the VRPDO, LNS-SPP competes with state-of-the-art algorithms on instances with up to 200 customers. On the 400 customers instances, the LNS-SPP would probably benefit from more tailored operators and local searches.

Table 6
Summary of the results on the VRPRDL instances of Reyes et al. (2017) and Ozbaygin et al. (2017).

		B&P		LNS-SPP						
				Fixed nbRoutes		Average of 5		Best of 5		
Instances	|N|	Routes	Cost*	Avg. cost	Best cost	Routes	Cost.0	Routes	Cost	Time (in s)
1–5	15	24	6 072*	6 072.0	6 072	22.0	6 119.0	22	6 119	3
6–10	20	28	6 848*	6 848.0	6 848	27.0	6 848.0	27	6 848	4
11–20	30	68	18 595*	18 595.0	18 595	67.0	18 651.4	67	18 639	8
21–30	60	129	37 213*	37 213.0	37 213	127.0	37 535.0	127	37 535	12
31–40	120	195	53 881*	53 738.4	53 738	178.8	53 826.0	178	53 918	60
41-50_v1	40	94	29 842*	29 838.0	29 838	93.0	29 855.4	93	29 838	10
41-50_v2	40	75	21 863*	21 864.4	21 863	71.0	21 928.4	71	21 927	10
Total		613	174 314	174 168.8	174 167	585.8	174 763.2	585	174 824	

Table 7
Summary of the results on the VRPHRDL instances of Reyes et al. (2017) and Ozbaygin et al. (2017).

		B&P		LNS-SPP						
				Fixed nbRoutes		Average of 5		Best of 5		
Instances	|N|	Routes	Cost*	Avg. cost	Best cost	Routes	Cost.0	Routes	Cost	Time (in s)
1–5	15	19	5 450*	5 450.0	5 450	19.0	5 450	19	5 450	3
6–10	20	20	5 604*	5 604.0	5 604	20.0	5 604	20	5 604	4
11–20	30	52	15 128*	15 128.0	15 128	51.0	15 228	51	15 212	8
21–30	60	83	26 829*	26 800.0	26 800	83.0	26 800	83	26 800	12
31–40	120	132	38 610*	37 310.4	37 252	115.6	37 423	115	37 373	60
41-50_v1	40	88	27 997*	27 996.0	27 996	87.0	27 996	87	27 996	10
41-50_v2	40	67	20 977*	20 958.0	20 958	67.6	20 958	67	20 958	10
Total		461	142 595	139 259.2	139 188	443.2	139 459	442	139 393	

5.1.2. Results on the VRP with (home and) roaming delivery locations (VRPRDL / VRPHRDL)

The LNS-SPP was evaluated on 120 benchmark instances of the VRP with roaming delivery locations proposed by Reyes et al. (2017) and Ozbaygin et al. (2017). Two versions of the set of instances are used. In the VRPRDL, the time windows associated with the options of each customer are disjointed, and only trunk deliveries are considered. In the VRPHRDL, the first option of each customer is considered to be the home option and it has no time window. The others options remain unchanged. Instances 1 to 40 come from Reyes et al. (2017), but were modified by Ozbaygin et al. (2017) to satisfy the triangle inequality. Instances 41 to 50, v1 and v2, come from Ozbaygin et al. (2017).

Tables 6 and 7 compare the results obtained by the B&P of Ozbaygin et al. (2017) with those obtained with LNS-SPP. Each line represents a group of instances, whose name and number of customers are written in columns 1 and 2. The last line is the summation of all results over all instances. Columns 3 reports the total number of routes obtained by Ozbaygin et al. (2017) for all instances of each group. Column 4 indicates the corresponding routing costs. A "*" symbol indicates that optimality could not be proved for all instance in the corresponding group.

The results of LNS-SPP are summarized in the remaining columns, and are based on five runs on each instance. The model in Ozbaygin et al. (2017) minimizes the routing cost but does not minimize the number of vehicles. In order to have a fair comparison, the fleet size was first set at the value found in Ozbaygin et al. (2017) and then the routing costs were minimized. The corresponding average costs over 5 runs and best costs are reported in columns 5 and 6, respectively. Note that using an imposed number of vehicles may cause the LNS-SPP to return some empty routes. Then, the number of routes was minimized first, and then the routing costs. Columns 7 and 8 give the average number of routes and routing costs found by the LNS-SPP over 5 runs, respectively. Columns 9 and 10 give the best number of routes and routing costs over 5 runs, respectively. The last column indicates the time budget allocated to the LNS-SPP algorithm for solving each instance. This time budget has been empirically set such that the results obtained by LNS-SPP stabilize. We recall that Ozbaygin et al. (2017) set a time limit of two hours, which is reached for most instances in the group 31–40.

The detailed results for each instance are reported in Tables A.12, A.13, A.14 and A.15 in Appendix A.

The B&P results given in column 4 are proven optimal on 93 instances out of 120. With the same number of routes, the cost of the best solution of LNS-SPP (column 6) finds 97 similar results and improves the results on 23 instances. When the number of vehicles is minimized first (column 9 and 10), the number of vehicles is reduced, which results in a higher cost, on 12 instances. On these instances, the total number of vehicles is decreased from 128 to 110 (14%) and the routing costs increase from 33 543 to 34 250 (+2.1%), knowing that 10 of these 12 instances have a cost proven optimal. In addition, on

Table 8
Classes of VRPDO instances.

Instance type	Capacity	Avg. number of options per customer	Time windows width	
			Individual locations	Shared Locations
U	medium	2	2 to 6 hours	8 to 12 hours
UBC	big	2	2 to 6 hours	8 to 12 hours
V	medium	1.5	2 to 6 hours	8 to 12 hours

Table 9
Summary of the results obtained by LNS-SPP on the VRPDO instances.

				Average of 5		Best of 5		
Type	$\|N\|$	$\|L\|$	$\|O\|$	Routes	Cost.0	Routes	Cost.0	Time (s)
U	50	60	100	54.0	3 913.88	54	3 864.48	30
	100	120	200	105.0	6 577.74	105	6 502.99	90
	200	240	400	205.0	14 082.36	205	13 641.13	300
	400	480	800	406.0	25 678.86	406	25 149.35	2000
UBC	50	52	100	20.0	2 301.31	20	2 293.53	30
	100	104	200	40.0	3 666.90	40	3 608.39	90
	200	209	400	80.0	6 534.72	80	6 384.29	300
	400	416	800	156.0	12 991.69	156	12 668.90	2000
V	50	48	75	54.0	3 759.76	54	3 742.59	30
	100	96	150	104.6	7 172.94	104	7 089.06	90
	200	192	300	205.0	15 261.14	205	14 781.23	300
	400	384	600	405.0	25 112.21	405	24 843.84	2000
Total				1 834.6	126 469.66	1 833	124 537.55	

3 instance, the number of vehicles is reduced and the cost is similar. Finally, in 14 instances both the number of vehicles and the cost were improved.

Unfortunately, no fair comparison can be made with the metaheuristic of Reyes et al. (2017) because the instances were modified by Ozbaygin et al. (2017). Furthermore, running times are not indicated in Reyes et al. (2017). Their method does not succeed in finding optimal solutions for small instances with 15 and 20 customers (with respect to optimal solutions provided by Gurobi 5.6).

The results on the VRPRDL and VRPHRDL show that the proposed LNS-SPP performs well on problems with delivery options, even with a limited time budget.

5.2. Instances of the VRPDO

VRPDO instances were randomly generated, because we did not find any suitable existing instances. Three types of instances were generated: U, V and UBC. For each type, instances with 50, 100, 200 and 400 customers were generated.

All the delivery locations were randomly generated in a 50×50 square. The depot is located at the bottom left-hand corner. Euclidean distances are considered. A unit of distance costs 1 and takes a unit of time to be crossed. A time horizon of 12 hours, i.e. 720 time units, is considered.

In the U and UBC instances, each customer has between 1 and 3 options, with an average of 2 options per customer. In the V instance, each customer has 1 or 2 options, with an average of 1.5 options per customer.

In the U and V instances, the capacities are tight, both for the vehicles and the lockers. A locker can accept between 3 and 5 parcels, and vehicle capacity is such that a route can serve around 10 customers. In the UBC instances (U with Big Capacity), the vehicle capacity is such that a route can serve around 25 customers. Furthermore, there are five times fewer lockers and their capacity is five times larger.

The time window of individual locations can be either: the morning ($[0; 360]$), the afternoon ($[360; 720]$), random in the morning (i.e $[a_i, b_i]$ such that $0 \leqslant a_i \leqslant 240$ and $b_i = a_i + 120$), random in the afternoon (i.e $[a_i, b_i]$ such that $360 \leqslant a_i \leqslant 600$ and $b_i = a_i + 120$) or random in the whole day (i.e $[a_i, b_i]$ such that $0 \leqslant a_i \leqslant 480$ and $b_i = a_i + 240$). The time window of a shared location can be either: random in the day (i.e $[a_i, b_i]$ such that $0 \leqslant a_i \leqslant 240$ and $b_i = a_i + 480$) or the full day ($[0; 720]$).

The characteristics of instance classes are summarized in Table 8. For each size and each class, 10 instances were generated, leading to a total of 120 instances. All the instances are available upon request.

Table 9 summarizes the results obtained by LNS-SPP on these instances. Each line represents a group of instances. The type of instance, the number of customers, the number of delivery locations and the number of options are specified in columns 1, 2, 3 and 4, respectively. Columns 5 and 6 depict the number of vehicles and the sum of average routing costs on

Table 10
Economic impact of delivery options. This table compares the results of the VRPTW, where the only option considered is home, with those of the VRPDO, including individual and shared delivery locations (SDL)

		VRPTW		VRPDO				
Type	$\|N\|$	Routes	Cost	Routes	Cost	Gap (%)	% SDL	SDL fill rate
U	50	54	5 504.07	54	3 864.48	29.8	58%	67%
	100	105	8 662.22	105	6 502.99	24.9	58%	67%
	200	205	16 223.30	205	13 641.13	15.9	60%	67%
UBC	50	20	5 253.71	20	2 293.53	56.3	67%	71%
	100	40	7 052.06	40	3 608.39	48.8	68%	67%
	200	80	10 681.93	80	6 384.29	40.2	66%	64%
V	50	54	5 192.10	54	3 742.59	27.9	57%	72%
	100	104	9 877.77	104	7 089.06	28.2	56%	79%
	200	205	18 945.38	205	14 781.23	22.0	54%	78%
Total		867	87 392.53	867	61 907.68	29.2		

all instances of the class considered, over 5 runs. Columns 7 and 8 depict the number of vehicles and the sum of the routing costs over on all instances of the class considered, keeping the best solution over 5 runs. Column 9 gives the time budget given for each run, for all instances of the class considered. The detailed results for each instance are given in Tables B.16, B.17 and B.18 in Appendix B.

By default, all these tests were conducted with a service level of 80% – 90%, i.e. at least 80% of the customers are served with their preferred option and at least 90% of the customers are served with an option of level 1 or 2.

A Wilcoxon pairwise test was performed in order to check the relevance of the SPP component. For instances with 50 and 100 customers, the Wilcoxon pairwise test concludes that LNS and LNS-SPP are not significantly different, with a p-value of 0.29. When all instances are considered, the results are significantly different with a p-value of 0.007. Indeed, removing the SPP component deteriorates the total routing cost by 2.26% on average, the number of vehicles remains the same.

5.3. Managerial insights for the VRPDO

5.3.1. Economic impact of delivery options

To quantify the impact of delivery options, the solution of the VRPDO instances with their VRPTW counterpart are compared. To transform a VRPDO instance into a VRPTW instance, only the home option is considered. No preference level is taken into account. In the VRPDO instances, home delivery options are assumed to be the preferred individual locations. In the case of customers that only have a locker option, a random location is added as a home location.

The total number of vehicles and the routing cost for each instance class are depicted in Table 10.

Each line represents the average results over the 10 instances of each group. The first column indicates the type of instance and the second column the number of customers. Columns 3 and 4 indicate the total number of vehicles and the sum of routing costs over the instances of the group when only home delivery is considered. Columns 5 and 6 indicate the total number of vehicles and the total routing cost over the instances of the group when all options are considered. Column 7 is the relative savings on the routing costs obtained by using delivery options. Considering delivery options leads to a cost reduction of 29.2%, on average. Furthermore, on the UBC instances, with large lockers, the savings are even larger. The number of routes is not reduced, because it is determined by the binding vehicles' capacities.

The use of SDLs is detailed in the last two columns, which detail the percentage of customers delivered at SDLs and the average fill rate of SDLs, respectively. These results indicate that the value of the gap in column 7 is related to the use of the shared locations. It is noticeable that a larger part of customers are delivered at SDLs in UBC instances with big capacities. But the last column shows that SDLs are not saturated. This suggests that alternative individual locations play a non negligible role in the gap between the VRPTW and the VRPDO.

Additionally, we checked the actual service levels on a representative set of solutions and observed that the percentage of customers served with their option of level 1 (resp. level 2) is very close to the thresholds of 80% and 90%. The service level constraints in the model are often binding.

5.3.2. Delivery options are compatible with tight time windows

We performed a sensitivity analysis by modifying the width of individual locations' time windows. Considering a time window $[a_i, b_i]$ at location $i \in L$, the modified time window is still centered at time $(a_i + b_i)/2$ but its width is reduced by a factor α as shown in formula (22).

$$[a'_i, b'_i] = \left[\frac{a_i + b_i}{2} - \frac{b_i - a_i}{2\alpha} \; ; \; \frac{a_i + b_i}{2} + \frac{b_i - a_i}{2\alpha} \right]. \tag{22}$$

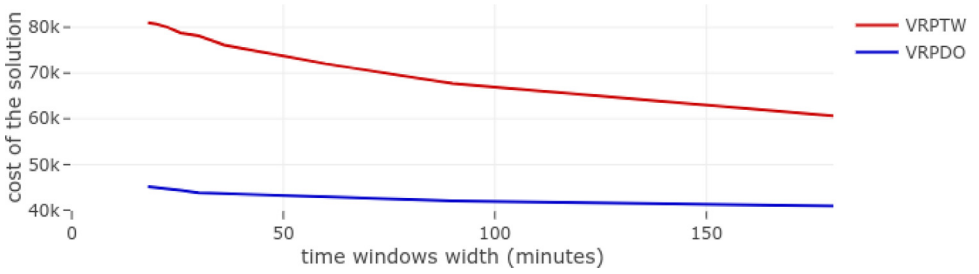


Fig. 12. Impact on time windows on the routing costs.

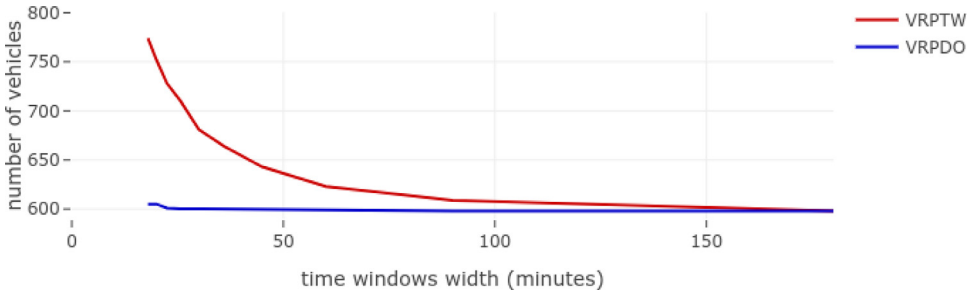


Fig. 13. Impact of time windows on the fleet size.

Table 11
Impact of service level on the routing costs

		% of customers served with option 1 (β_1)		
		70%	80%	90%
% of customers served with options 1 or 2	80	60 215.29	61 475.93*	-*
	90	60 393.85	61 907.68*	66 155.67*
	95	60 710.38	61 980.41*	66 574.68*
	100	61 892.21	63 172.80*	-

The same experiments as in Table 10 were conducted with these tighter time windows. Fig. 12 represents the routing costs of the solutions of the VRPTW (red line) and the VRPDO (blue line), respectively, when the width of time windows at individual locations varies.

The x-axis represent the width of time windows and the y-axis shows the routing costs The time windows width was set by applying a factor α going from 1 and 10 in equation (22), leading to time window width between 20 and 180 minutes. Note that some instances become infeasible when the time windows are too tight. Thus, not all instances are taken into account in this figure. The main conclusion of this figure is that the cost of both the VRPTW and the VRPDO increase when time windows are tightened, but this increase is only 10% in the case of the VRPDO.

Fig. 13 represents the number of vehicles used in the solution of the VRPTW (red line) and the VRPDO (blue line), respectively, when the width of time windows at individual locations varies between 20 and 180 minutes.

With large time windows, each route can deliver many customers, so that the bottleneck is the capacity of truck. This is the solutions of both the VRPTW and the VRPDO use 598 vehicles. With very tight time window, the number of vehicles needed in the VRPTW grows dramatically, while it increases from 598 to 604 for the VRPDO. The main conclusion is that considering delivery options and SDLs makes it possible to serve customers at individual delivery locations with very narrow time windows, at the cost of minor increase of the vehicle fleet and very reasonable cost increase.

To illustrate this property, let us consider instance U_100_1, with 100 customers, small trucks, and many small lockers. When the widths of original time windows are divided by 10, the number of vehicles remains the same, with an average fill rate of 90%. The routing costs increase by only 2.8%. A first explanation is that the number of customers served at an SDL is 65 and 66, respectively, and these locations are not impacted by the time window reduction. A second explanation is that 14 customers are delivered at another location and in another time window in both cases. Finally, note that the general service level constraints are binding in the two scenarios.

5.3.3. Delivery options and service level constraints

In order to quantify the impact of service level constraints, different values of parameters β_1 and β_2 were tested. Table 11 presents sum of the routing cost of all VRPDO instances, with respect to the required service level. Columns 3, 4 and 5 correspond to solutions i which the ratio of customers served with their preferred option (β_1) is at least 70, 80 and 90,

respectively. Lines 3 to 6 correspond to the solutions where the ration of customers delivered with options 1 or 2 (β_2) is equal to at least 80, 90, 95 and 100, respectively.

In most solutions of Table 11, the total number of vehicles used is 866. Sometimes, one additional vehicle is necessary ; in these cases, the routing costs are followed by a *. The cell corresponding to $(\beta_1, \beta_2) = (90, 80)$ is left empty because the corresponding values of β_1 and β_2 are incompatible. The cell of $(\beta_1, \beta_2) = (90, 100)$ is left empty because some instances are infeasible with such restrictive parameters.

This table shows that the service level can be greatly improved with no major increase of the routing costs. Moreover, the service level has very limited impact on the fleet size. The impact of β_2 seems to be rather limited. Indeed, the average cost increase when β_2 varies from 80 to 100 (with $\beta_1 = 70$) is only 2.8%, whereas the cost increase when β_1 varies from 70 to 90 (with $\beta_2 = 90$) is 9.5%.

6. Conclusion

In this paper we have proposed a new extension of the VRPTW allowing multiple delivery options for each customer. The Vehicle Routing Problem with Delivery Options (VRPDO) includes the use of shared delivery locations, such as lockers, and takes in account customer preferences. In addition, the model is general enough to include new modes of delivery, such as trunk delivery. The VRPDO is theoretically challenging because it introduces synchronized resources and a new structure of the search space, due to the presence of several delivery locations for each customer.

The numerical experiments show that VRPDO can save considerable amounts of money compared with VRPTW. Moreover, for a small cost increase, a very high quality of service can be achieved, especially with respect to the time window width.

The VRPDO is solved with an LNS meta-heuristic combined with a set partitioning component. After implementing a large number of ruin and recreate operators, a comprehensive tuning process resulted in the selection of a few relevant operators. The experiments show that combining local impact fast operators and global impact slower operators is an efficient strategy. Alternating between large and small removal operators help perform intensification as well as diversification.

For future research, we plan to use dynamic programming, as in Moccia et al. (2012) and Reyes et al. (2017), to improve promising solutions. The difficulty in efficiently applying these methods to the VRPDO is the combinatorial explosion of the number of labels induced by the synchronized resources. Another perspective is to integrate delivery options into multi-echelon (Grangier et al., 2017b) or multi-modal (Masson et al., 2017) city logistics systems, or to combine it with the use of autonomous vehicles (Boysen et al., 2018).

Author Statement

This paper discusses the Vehicle Routing Problem with Delivery Options (VRPDO). It is an extension of both the Vehicle Routing Problem with Time Windows (VRPTW) and the Generalized Vehicle Routing Problem (GVRP), in which customer can be served at various locations and service times.

This rising practice in parcel distribution will continue to develop a lot in future years. Similar application may also concern freight distribution, with the goal of optimizing the vehicles fill rate. In practice, each postal company can have their own implementation of such services, in particular there are several ways to assess the quality of service to customers. But the core of the problem is still an extension of the GVRP with distinct time windows.

We are aware of several research project in the same field, but, to the best of our knowledge, papers dealing with the VRP with delivery options have not yet been published in peer-reviewed journals.

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Appendix A. Detailed results on the VRPRDL and VRPHRDL instances

Tables A.12–A.15.

Table A1
Comparison with the results of Ozbaygin et al. (2017) on the VRPRDL instances of Reyes et al. (2017).

		B&P			LNS-SPP						
					Fixed nbRoutes		Average of 5		Best of 5		
Instances	Customers	Routes	Cost		Avg. cost	Best of 5 cost	Routes	Cost	Routes	Cost	Time (s)
1	15	4	901		901	901	4	901	4	901	3
2	15	5	1 286		1 286	1 286	5	1 286	5	1 286	3
3	15	4	991		991	991	3	999	3	999	3
4	15	5	1 062		1 062	1 062	4	1 101	4	1 101	3
5	15	6	1 832		1 832	1 832	6	1 832	6	1 832	3
Group avg		4.8	1 214.4		1 214.4	1 214.4	4.4	1 223.8	4.4	1 223.8	
6	20	5	1 294		1 294	1 294	5	1 294	5	1 294	4
7	20	4	1 155		1 155	1 155	4	1 155	4	1 155	4
8	20	6	1 455		1 455	1 455	6	1 455	6	1 455	4
9	20	5	1 260		1 260	1 260	5	1 260	5	1 260	4
10	20	8	1 684		1 684	1 684	7	1 684	7	1 684	4
Group avg		5.6	1 369.6		1 369.6	1 369.6	5.4	1 369.6	5.4	1 369.6	
11	30	7	1 922		1 922	1 922	7	1 922	7	1 922	8
12	30	8	2 324		2 324	2 324	8	2 324	8	2 324	8
13	30	6	1 747		1 747	1 747	6	1 747	6	1 747	8
14	30	6	1 273		1 273	1 273	5	1 329.4	5	1 317	8
15	30	6	1 694		1 694	1 694	6	1 694	6	1 694	8
16	30	7	1 938		1 938	1 938	7	1 938	7	1 938	8
17	30	8	1 965		1 965	1 965	8	1 965	8	1 965	8
18	30	7	1 827		1 827	1 827	7	1 827	7	1 827	8
19	30	7	2 083		2 083	2 083	7	2 083	7	2 083	8
20	30	6	1 822		1 822	1 822	6	1 822	6	1 822	8
Group avg		6.8	1 859.5		1 859.5	1 859.5	6.7	1 865.1	6.7	1 863.9	
21	60	13	3 761		3 761	3 761	13	3 761	13	3 761	12
22	60	10	2 828		2 828	2 828	10	2 828	10	2 828	12
23	60	16	4 440		4 440	4 440	16	4 440	16	4 440	12
24	60	11	3 378		3 378	3 378	11	3 378	11	3 378	12
25	60	11	3 161		3 161	3 161	11	3 161	11	3 161	12
26	60	16	4 536		4 536	4 536	16	4 536	16	4 536	12
27	60	10	2 865		2 865	2 865	9	2 976	9	2 976	12
28	60	14	4 173		4 173	4 173	14	4 173	14	4 173	12
29	60	14	3 964		3 964	3 964	13	4 175	13	4 175	12
30	60	14	4 107		4 107	4 107	14	4 107	14	4 107	12
Group avg		12.9	3 721.3		3 721.3	3 721.3	12.7	3 753.5	12.7	3 753.5	
31	120	18	4 935		4 935	4 935	16	4 938	16	4 938	60
32	120	19	5 278	*	5 258	5 258	17	5 273.4	17	5 263	60
33	120	18	5 083	*	5 061	5 061	17	5 061	17	5 061	60
34	120	17	5 218		5 218	5 218	17	5 218	17	5 218	60
35	120	20	5 519	*	5 498	5 498	18	5 528.6	18	5 526	60
36	120	22	6 498		6 498	6 498	21	6 498	21	6 498	60
37	120	17	4 845	*	4 830	4 830	17	4 830	17	4 830	60
38	120	21	5 608	*	5 604	5 604	19	5 604	19	5 604	60
39	120	24	5 849	*	5 841	5 841	19.8	5 869.8	19	5 985	60
40	120	19	5 048	*	4 995.4	4 995	17	5 005.2	17	4 995	60
Group avg		19.5	5 388.1		5 373.8	5 373.8	17.9	5 382.6	17.8	5 391.8	
Overall avg		11.1	3 065.2		3 061.6	3 061.6	10.6	3 074.5	10.5	3 076.5	

Table A2
Comparison with the results of Ozbaygin et al. (2017) on the VRPHRDL instances of Reyes et al. (2017).

		B&P			LNS-SPP						
					Fixed nbRoutes		Average of 5		Best of 5		
Instances	Customers	Routes	Cost		Avg. cost	Best of 5 cost	Routes	Cost	Routes	Cost	Time (s)
1	15	3	773		773	773	3	773	3	773	3
2	15	4	1 065		1 065	1 065	4	1 065	4	1 065	3
3	15	3	988		988	988	3	988	3	988	3
4	15	3	914		914	914	3	914	3	914	3
5	15	6	1 710		1 710	1 710	6	1 710	6	1 710	3
Group avg		3.8	1 090.0		1 090.0	1 090.0	3.8	1 090	3.8	1 090	
6	20	4	1 099		1 099	1 099	4	1 099	4	1 099	4
7	20	3	996		996	996	3	996	3	996	4
8	20	5	1 346		1 346	1 346	5	1 346	5	1 346	4
9	20	4	997		997	997	4	997	4	997	4
10	20	4	1 166		1 166	1 166	4	1 166	4	1 166	4
Group avg		4.0	1 120.8		1 120.8	1 120.8	4	1 120.8	4	1 120.8	
11	30	5	1 587		1 587	1 587	5	1 593.6	5	1 587	8
12	30	6	1 808		1 808	1 808	6	1 808	6	1 808	8
13	30	6	1 563		1 563	1 563	5	1 647	5	1 647	8
14	30	4	1 058		1 058	1 058	4	1 058	4	1 058	8
15	30	5	1 347		1 347	1 347	5	1 347	5	1 347	8
16	30	5	1 517	*	1 517	1 517	5	1 517	5	1 517	8
17	30	5	1 445		1 445	1 445	5	1 445	5	1 445	8
18	30	5	1 627		1 627	1 627	5	1 636.4	5	1 627	8
19	30	5	1 461		1 461	1 461	5	1 461	5	1 461	8
20	30	6	1 715		1 715	1 715	6	1 715	6	1 715	8
Group avg		5.2	1 512.8		1 512.8	1 512.8	5.1	1 522.8	5.1	1 521.2	
21	60	8	2 580		2 580	2 580	8	2 580	8	2 580	12
22	60	7	2 213	*	2 206	2 206	7	2 206	7	2 206	12
23	60	10	3 363		3 363	3 363	10	3 363	10	3 363	12
24	60	8	2 569	*	2 569	2 569	8	2 569	8	2 569	12
25	60	8	2 400	*	2 378	2 378	8	2 378	8	2 378	12
26	60	9	2 845	*	2 845	2 845	9	2 845	9	2 845	12
27	60	8	2 518		2 518	2 518	8	2 518	8	2 518	12
28	60	8	2 758		2 758	2 758	8	2 758	8	2 758	12
29	60	9	2 892	*	2 892	2 892	9	2 892	9	2 892	12
30	60	8	2 691		2 691	2 691	8	2 691	8	2 691	12
Group avg		8.3	2 682.9		2680.0	2 680.0	8.3	2 680.0	8.3	2 680.0	
31	120	14	3 984	*	3 666	3 666	11	3 666	11	3 666	60
32	120	13	3 958	*	3 885	3 885	12	3 886.4	12	3 885	60
33	120	13	3 630	*	3 543.6	3 543	11.4	3 567	11	3 544	60
34	120	13	3 891	*	3 711.8	3 694	12	3 784.8	12	3 783	60
35	120	11	3 255	*	3 184	3 184	10	3 184	10	3 184	60
36	120	15	4 525	*	4 311	4 273	13.2	4 326	13	4 304	60
37	120	11	3 395	*	3 217	3 217	10	3 217	10	3 217	60
38	120	14	3 976	*	3 935.8	3 935	12	3 935	12	3 935	60
39	120	15	4 316	*	4 300	4 300	13	4 300	13	4 300	60
40	120	13	3 680	*	3 556.2	3 555	11	3 556.8	11	3 555	60
Group avg		13.2	3 861.0		3 731.0	3 725.2	11.6	3 742.3	11.5	3 737.3	
Overall avg		7.6	2 290.5		2 257.3	2 255.8	7.2	2 262.6	7.2	2 261	

Table A3
Comparison with the results of Ozbaygin et al. (2017) on the VRPRDL instances of Ozbaygin et al. (2017).

	B&P			LNS-SPP						
				Fixed nbRoutes		Average of 5		Best of 5		
Instances	Routes	Cost		Avg. cost	Best of 5 cost	Routes	Cost	Routes	Cost	Time (s)
41_v1	10	3 203		3 203	3 203	10	3 219.4	10	3 203	10
41_v2	7	2 133		2 134.4	2 133	7	2 134.4	7	2 133	10
42_v1	9	2 799		2 799	2 799	9	2 799	9	2 799	10
42_v2	7	1 946		1 946	1 946	6	1 946	6	1 946	10
43_v1	8	2 607	*	2 603	2 603	8	2 604	8	2 603	10
43_v2	8	1 966		1 966	1 966	7	1 967	7	1 967	10
44_v1	7	2 261		2 261	2 261	7	2 261	7	2 261	10
44_v2	6	1 610		1 610	1 610	5	1 614	5	1 614	10
45_v1	10	3 217		3 217	3 217	10	3 217	10	3 217	10
45_v2	8	2 478		2 478	2 478	8	2 478	8	2 478	10
46_v1	9	2 805		2 805	2 805	9	2 805	9	2 805	10
46_v2	8	2 469		2 469	2 469	8	2 469	8	2 469	10
47_v1	10	3 339		3 339	3 339	10	3 339	10	3 339	10
47_v2	7	1 946		1 946	1 946	6	2 005	6	2 005	10
48_v1	10	3 325		3 325	3 325	10	3 325	10	3 325	10
48_v2	8	2 380		2 380	2 380	8	2 380	8	2 380	10
49_v1	11	3 534		3 534	3 534	11	3 534	11	3 534	10
49_v2	8	2 492		2 492	2 492	8	2 492	8	2 492	10
50_v1	10	2 752		2 752	2 752	9	2 752	9	2 752	10
50_v2	8	2 443		2 443	2 443	8	2 443	8	2 443	10
Avg	8.4	2 585.2		2 5851	2 585.0	8.2	2 589.2	8.2	2 588.2	

Table A4
Comparison with the results of Ozbaygin et al. (2017) on the VRPHRDL instances of Ozbaygin et al. (2017).

	B&P			LNS-SPP						
				Fixed nbRoutes		Average of 5		Best of 5		
Instances	Routes	Cost		Avg. cost	Best of 5 cost	Routes	Cost	Routes	Cost	Time (s)
41_v1	8	2 662		2 662	2 662	8	2 662	8	2 662	10
41_v2	6	1 998		1 998	1 998	6.6	1 998	6	1 998	10
42_v1	8	2 610		2 610	2 610	8	2 610	8	2 610	10
42_v2	6	1 946	*	1 927	1 927	6	1 927	6	1 927	10
43_v1	7	2 260		2 260	2 260	7	2 260	7	2 260	10
43_v2	6	1 830		1 830	1 830	6	1 830	6	1 830	10
44_v1	7	2 147		2 147	2 147	7	2 147	7	2 147	10
44_v2	5	1 478		1 478	1 478	5	1 478	5	1 478	10
45_v1	10	3 172		3 172	3 172	10	3 172	10	3 172	10
45_v2	8	2 466		2 466	2 466	8	2 466	8	2 466	10
46_v1	8	2 616		2 616	1 616	8	2 616	8	2 616	10
46_v2	8	2 388		2 388	2 388	8	2 388	8	2 388	10
47_v1	9	3 011	*	3 010	3 010	9	3 010	9	3 010	10
47_v2	6	1 848		1 848	1 848	6	1 848	6	1 848	10
48_v1	10	3 278		3 278	3 278	10	3 278	10	3 278	10
48_v2	7	2 264		2 264	2 264	7	2 264	7	2 264	10
49_v1	11	3 514	*	3 514	3 514	11	3 514	11	3 514	10
49_v2	8	2 457		2 457	2 457	8	2 457	8	2 457	10
50_v1	10	2 727		2 727	2 727	9	2 727	9	2 727	10
50_v2	7	2 302		2 302	2 302	7	2 302	7	2 302	10
Avg	7.7	2 448.7		2 447.7	2 447.7	7.7	2 447.7	7.7	2 447.7	

Appendix B. Detailed results on the VRPDO instances

Tables B1–B3.

Table B1
Detailed results on the U instances of the VRPDO.

			Average of 5		Best of 5	
Instance	Customers	Time (s)	Routes	Cost.0	Routes	Cost.0
U_50_1	50	30	6	434.773	6	433.081
U_50_2	50	30	6	424.665	6	423.228
U_50_3	50	30	5	329.138	6	329.138
U_50_4	50	30	5	448.321	5	427.183
U_50_5	50	30	6	354.322	6	353.179
U_50_6	50	30	5	403.979	5	399.620
U_50_7	50	30	5	336.428	5	332.481
U_50_8	50	30	5	387.648	5	384.635
U_50_9	50	30	5	360.930	5	359.888
U_50_10	50	30	6	433.679	6	422.048
Total			54	3 913.883	54	3 864.481
U_100_1	100	90	11	484.226	11	479.972
U_100_2	100	90	10	687.754	10	683.365
U_100_3	100	90	11	634.604	11	630.215
U_100_4	100	90	10	610.063	10	592.860
U_100_5	100	90	10	678.523	10	673.844
U_100_6	100	90	10	597.610	10	586.810
U_100_7	100	90	11	761.344	11	746.400
U_100_8	100	90	11	856.017	11	847.533
U_100_9	100	90	10	690.985	10	687.659
U_100_10	100	90	11	576.614	11	574.328
Total			105	6 577.739	105	6 502.986
U_200_1	200	300	21	1 706.754	21	1 687.420
U_200_2	200	300	21	1 326.250	21	1 287.250
U_200_3	200	300	20	1 603.076	20	1 565.130
U_200_4	200	300	21	968.057	21	943.988
U_200_5	200	300	21	1 164.854	21	1 112.500
U_200_6	200	300	20	1 310.448	20	1 261.360
U_200_7	200	300	21	1 166.970	21	1 108.010
U_200_8	200	300	20	1 345.542	20	1 259.660
U_200_9	200	300	20	1 623.972	20	1 575.460
U_200_10	200	300	20	1 866.434	20	1 840.350
Total			205	14 082.357	205	13 641.128
U_400_1	400	2000	41	2 039.058	41	2 014.570
U_400_2	400	2000	41	3 663.066	41	3 614.200
U_400_3	400	2000	41	2 686.832	41	2 641.320
U_400_4	400	2000	40	2 008.818	40	1 971.310
U_400_5	400	2000	40	2 472.460	40	2 437.530
U_400_6	400	2000	40	2 167.936	40	2 145.120
U_400_7	400	2000	41	2 972.060	41	2 909.060
U_400_8	400	2000	41	2 065.766	41	2 032.460
U_400_9	400	2000	40	3 152.348	40	3 058.030
U_400_10	400	2000	41	2 450.518	41	2 325.750
Total			406	25 678.862	406	25 149.350

Table B2
Detailed results on the UBC instances of the VRPDO.

			Average of 5		Best of 5	
Instance	Customers	Time (s)	Routes	Cost.0	Routes	Cost.0
UBC_50_1	50	30	2	256.042	2	256.042
UBC_50_2	50	30	2	305.678	2	301.963
UBC_50_3	50	30	2	241.696	2	241.696
UBC_50_4	50	30	2	226.627	2	223.522
UBC_50_5	50	30	2	208.172	2	208.096
UBC_50_6	50	30	2	242.484	2	241.846
UBC_50_7	50	30	2	246.186	2	246.186
UBC_50_8	50	30	2	196.079	2	196.079
UBC_50_9	50	30	2	184.054	2	183.804
UBC_50_10	50	30	2	194.297	2	194.297
Total			20	2 301.314	20	2 293.531
UBC_100_1	100	90	4	383.743	4	374.862
UBC_100_2	100	90	4	357.865	4	351.220
UBC_100_3	100	90	4	330.514	4	323.464
UBC_100_4	100	90	4	335.395	4	334.615
UBC_100_5	100	90	4	376.359	4	371.859
UBC_100_6	100	90	4	364.341	4	357.611
UBC_100_7	100	90	4	339.716	4	337.902
UBC_100_8	100	90	4	426.310	4	419.714
UBC_100_9	100	90	4	394.706	4	386.871
UBC_100_10	100	90	4	357.947	4	350.272
Total			40	3 666.896	40	3 608.39
UBC_200_1	200	300	8	609.565	8	601.353
UBC_200_2	200	300	8	525.744	8	511.301
UBC_200_3	200	300	8	851.712	8	842.696
UBC_200_4	200	300	8	670.960	8	634.515
UBC_200_5	200	300	8	674.374	8	669.268
UBC_200_6	200	300	8	665.084	8	649.461
UBC_200_7	200	300	8	649.339	8	630.953
UBC_200_8	200	300	8	655.330	8	640.022
UBC_200_9	200	300	8	582.294	8	562.377
UBC_200_10	200	300	8	650.315	8	562.377
Total			80	6 534.717	80	6 384.287
UBC_400_1	400	2000	16	1 044.096	16	1 019.140
UBC_400_2	400	2000	16	1 247.336	16	1 247.220
UBC_400_3	400	2000	15	1 498.534	15	1 433.390
UBC_400_4	400	2000	16	1 253.736	16	1 243.220
UBC_400_5	400	2000	16	1 083.622	16	1 045.200
UBC_400_6	400	2000	15	1 576.400	15	1 534.190
UBC_400_7	400	2000	15	1 151.348	15	1 111.400
UBC_400_8	400	2000	16	1 313.132	16	1 270.820
UBC_400_9	400	2000	16	1 237.992	16	1 231.080
UBC_400_10	400	2000	15	1 585.494	15	1 533.240
Total			156	12 991.690	156	12 668.900

Table B3
Detailed results on the V instances of the VRPDO.

			Average of 5		Best of 5	
Instance	Customers	Time (s)	Routes	Cost.0	Routes	Cost.0
V_50_1	50	30	5	417.903	5	416.349
V_50_2	50	30	6	345.635	6	345.635
V_50_3	50	30	5	401.676	5	398.662
V_50_4	50	30	5	378.415	5	374.501
V_50_5	50	30	5	328.692	5	328.387
V_50_6	50	30	5	390.432	5	387.196
V_50_7	50	30	5	368.295	5	365.985
V_50_8	50	30	6	387.645	6	387.645
V_50_9	50	30	6	366.000	6	366.000
V_50_10	50	30	6	375.068	6	372.228
Total			54	3 759.762	54	3 742.588
V_100_1	100	90	11	652.508	11	647.566
V_100_2	100	90	10.2	905.803	10	895.205
V_100_3	100	90	10.4	743.616	10	733.412
V_100_4	100	90	10	596.960	10	594.605
V_100_5	100	90	10	795.629	10	765.491
V_100_6	100	90	11	597.567	11	597.288
V_100_7	100	90	11	705.792	11	702.181
V_100_8	100	90	11	745.717	11	744.855
V_100_9	100	90	10	812.543	10	794.041
V_100_10	100	90	10	616.804	10	614.412
Total			104.6	7 172.939	104	7 089.056
V_200_1	200	300	21	1 321.538	21	1 288.280
V_200_2	200	300	20	1 526.820	20	1 420.190
V_200_3	200	300	21	1 369.850	21	1 346.210
V_200_4	200	300	21	1 545.578	21	1 488.720
V_200_5	200	300	21	1 445.074	21	1 420.810
V_200_6	200	300	21	1 562.392	21	1 519.580
V_200_7	200	300	20	1 204.788	20	1 171.590
V_200_8	200	300	20	1 851.056	20	1 799.830
V_200_9	200	300	20	1 837.234	20	1 798.400
V_200_10	200	300	20	1 596.814	20	1 527.620
Total			205	15 261.144	205	14 781.23
V_400_1	400	2000	41	2 798.576	41	2 763.730
V_400_2	400	2000	41	3 120.520	41	3 065.630
V_400_3	400	2000	40.4	2 110.876	40	2 118.930
V_400_4	400	2000	40	2 096.798	40	2 072.650
V_400_5	400	2000	41	1 772.150	41	1 759.510
V_400_6	400	2000	40	3 299.128	40	3 281.810
V_400_7	400	2000	41	2 393.084	41	2 391.090
V_400_8	400	2000	41	1 992.048	41	1 963.520
V_400_9	400	2000	40	2 488.052	40	2 448.350
V_400_10	400	2000	40	3 040.982	40	2 978.620
Total			405.4	25 112.214	205	24 843.840

References

Afsar, H.M., Prins, C., Santos, A.C., 2014. Exact and heuristic algorithms for solving the generalized vehicle routing problem with flexible fleet size. International Transactions in Operational Research 21 (1), 153–175. doi:10.1111/itor.12041.

Agatz, N., Campbell, A., Fleischmann, M., Savelsbergh, M., 2008. Time slot management in attended home delivery. Erasmus Research Institute of Management 45. doi:10.2307/23018537.

Allahyari, S., Salari, M., Vigo, D., 2015. A hybrid metaheuristic algorithm for the multi-depot covering tour vehicle routing problem. Eur J Oper Res 242 (3), 756–768. doi:10.1016/j.ejor.2014.10.048.

Allen, J., Piecyk, M., Piotrowska, M., 2016. Analysis of the Parcels Market and Parcel Carriers Operations in the UK. Technical Report. University of Westminster.

Archetti, C., Speranza, M.G., Vigo, D., 2014. Chapter 10: vehicle routing problems with profits. In: Vehicle Routing: Problems, Methods, and Applications, Second Edition. SIAM, pp. 273–297. doi:10.1137/1.9781611973594.ch10.

Belhaiza, S., Hansen, P., Laporte, G., 2014. A hybrid variable neighborhood tabu search heuristic for the vehicle routing problem with multiple time windows. Computers & Operations Research 52, 269–281. doi:10.1016/j.cor.2013.08.010.

Bernardino, R., Paias, A., 2018. Metaheuristics based on decision hierarchies for the traveling purchaser problem. International Transactions in Operational Research 25 (4), 1269–1295. doi:10.1111/itor.12330.

Boysen, N., Schwerdfeger, S., Weidinger, F., 2018. Scheduling last-mile deliveries with truck-based autonomous robots. Eur J Oper Res 271 (3), 1085–1099. doi:10.1016/j.ejor.2018.05.058.

Bräysy, O., Gendreau, M., 2005. Vehicle routing problem with time windows, part I: route construction and local search algorithms. Transportation Science 39 (1), 104–118. doi:10.1287/trsc.1030.0056.

Bräysy, O., Gendreau, M., 2005. Vehicle routing problem with time windows, part II: metaheuristics. Transportation Science 39 (1), 119–139. doi:10.1287/trsc.1030.0057.

Christiaens, J., Vanden Berghe, G., 2020. Slack induction by string removals for vehicle routing problems. Transportation Science.

Cordeau, J.-F., Desaulniers, G., Desrosiers, J., Solomon, M.M., Soumis, F., 2002. 7. VRP with time windows. In: The Vehicle Routing Problem. SIAM, pp. 157–193.

Curtois, T., Landa-Silva, D., Qu, Y., Laesanklang, W., 2018. Large neighbourhood search with adaptive guided ejection search for the pickup and delivery problem with time windows. EURO Journal on Transportation and Logistics 7 (2), 151–192. doi:10.1007/s13676-017-0115-6.

Demir, E., Bektaş, T., Laporte, G., 2012. An adaptive large neighborhood search heuristic for the pollution-routing problem. Eur J Oper Res 223 (2), 346–359. doi:10.1016/j.ejor.2012.06.044.

Drexl, M., 2012. Synchronization in vehicle routing-a survey of VRPs with multiple synchronization constraints. Transportation Science 46 (3), 297–316. doi:10.1287/trsc.1110.0400.

Dueck, G., 1993. New optimization heuristics: the great deluge algorithm and the record-to-record travel. J Comput Phys 104 (1), 86–92. doi:10.1006/jcph.1993.1010.

Favaretto, D., Moretti, E., Pellegrini, P., 2007. Ant colony system for a VRP with multiple time windows and multiple visits. Journal of Interdisciplinary Mathematics 10 (2), 263–284. doi:10.1080/09720502.2007.10700491.

Florio, A.M., Feillet, D., Hartl, R.F., 2018. The delivery problem: optimizing hit rates in e-commerce deliveries. Transportation Research Part B: Methodological 117, 455–472. doi:10.1016/j.trb.2018.09.011.

Foster, B.A., Ryan, D.M., 1976. An integer programming approach to the vehicle scheduling problem. Journal of the Operational Research Society 27 (2), 367–384. doi:10.1057/jors.1976.63.

Froger, A., Mendoza, J.E., Jabali, O., Laporte, G., 2017. A matheuristic for the electric vehicle routing problem with capacitated charging stations. Technical Report. CIRRELT.

Gehring, H., Homberger, J., 1999. A parallel hybrid evolutionary metaheuristic for the vehicle routing problem with time windows. In: Proceedings of EUROGEN99, 2, pp. 57–64.

Ghiani, G., Improta, G., 2000. An efficient transformation of the generalized vehicle routing problem. Eur J Oper Res 122 (1), 11–17. doi:10.1016/S0377-2217(99)00073-9.

Grangier, P., Gendreau, M., Lehuédé, F., Rousseau, L.-M., 2017. A matheuristic based on large neighborhood search for the vehicle routing problem with cross-docking. Computers & Operations Research 84, 116–126. doi:10.1016/j.cor.2017.03.004.

Grangier, P., Gendreau, M., Lehuédé, F., Rousseau, L.-M., 2017. The vehicle routing problem with cross-docking and resource constraints. Journal of Heuristics 1–31. doi:10.1007/s10732-019-09423-y.

Groër, C., Golden, B., Wasil, E., 2011. A parallel algorithm for the vehicle routing problem. INFORMS J Comput 23 (2), 315–330. doi:10.1023/A:1018948011707.

Gschwind, T., Drexl, M., 2019. Adaptive large neighborhood search with a constant-time feasibility test for the dial-a-ride problem. Transportation Science 53 (2), 480–491. doi:10.1287/trsc.2018.0837.

Hempsch, C., Irnich, S., 2008. Vehicle routing problems with inter-tour resource constraints. In: The Vehicle Routing Problem: Latest Advances and New Challenges. Springer, pp. 421–444. doi:10.1007/978-0-387-77778-8_19.

Hoogeboom, M., Dullaert, W., 2019. Vehicle routing with arrival time diversification. Eur J Oper Res 275, 93–107. doi:10.1016/j.ejor.2018.11.020.

Kammoun, M., Derbel, H., Ratli, M., Jarboui, B., 2017. An integration of mixed VND and VNS: the case of the multivehicle covering tour problem. International Transactions in Operational Research 24 (3), 663–679. doi:10.1111/itor.12355.

Kirkpatrick, S., Gelatt, C.D., Vecchi, M.P., 1983. Optimization by simulated annealing. Science 220 (4598), 671–680. doi:10.1126/science.220.4598.671.

López-Ibánez, M., Dubois-Lacoste, J., Stützle, T., Birattari, M., 2011. The IRACE package, iterated race for automatic algorithm configuration. Technical Report. Université Libre de Bruxelles.

Mancini, S., 2017. A combined multistart random constructive heuristic and set partitioning based formulation for the vehicle routing problem with time dependent travel times. Computers & Operations Research 88, 290–296. doi:10.1016/j.cor.2017.06.021.

Masson, R., Trentini, A., Lehuédé, F., Malhéné, N., Péton, O., Tlahig, H., 2017. Optimization of a city logistics transportation system with mixed passengers and goods. EURO Journal on Transportation and Logistics 6 (1), 81–109.

McFarland, M., 2018. Amazon Now Delivers to the Trunk of Your Car. CNN business.

Mendoza, J.E., Villegas, J.G., 2013. A multi-space sampling heuristic for the vehicle routing problem with stochastic demands. Optimization Letters 7 (7), 1503–1516. doi:10.1007/s11590-012-0555-8.

Moccia, L., Cordeau, J.-F., Laporte, G., 2012. An incremental tabu search heuristic for the generalized vehicle routing problem with time windows. Journal of the Operational Research Society 63 (2), 232–244. doi:10.1057/jors.2011.25.

Morganti, E., Seidel, S., Blanquart, C., Dablanc, L., Lenz, B., 2014. The impact of e-commerce on final deliveries: alternative parcel delivery services in France and Germany. Transp. Res. Procedia 4, 178–190. doi:10.1016/j.trpro.2014.11.014.

Moyou, E., 2019. Chiffre D'affaires Annuel Du E-commerce En France De 2005 2018.

Nagata, Y., Bräysy, O., 2009. A powerful route minimization heuristic for the vehicle routing problem with time windows. Operations Research Letters 37 (5), 333–338. doi:10.1016/j.orl.2009.04.006.

Olkis, K., Tilk, C., Irnich, S., 2020. The Last-mile Vehicle Routing Problem with Delivery Options. Working Paper. Gutenberg School of Management and Economics, Chair of Logistics Management, Johannes Gutenberg University, Mainz, Germany.

Ozbaygin, G., Karasan, O.E., Savelsbergh, M., Yaman, H., 2017. A branch-and-price algorithm for the vehicle routing problem with roaming delivery locations. Transportation Research Part B: Methodological 100, 115–137. doi:10.1016/j.trb.2017.02.003.

Parragh, S.N., Schmid, V., 2013. Hybrid column generation and large neighborhood search for the dial-a-ride problem. Computers & Operations Research 40 (1), 490–497. doi:10.1016/j.cor.2012.08.004.
Pisinger, D., Ropke, S., 2007. A general heuristic for vehicle routing problems. Computers and Operations Research 34 (8), 2403–2435. doi:10.1016/j.cor.2005.09.012.
Pisinger, D., Ropke, S., 2019. Large neighborhood search. In: Gendreau, M., Potvin, J.-Y. (Eds.), Handbook of Metaheuristics. Springer International Publishing, pp. 99–127.
Prescott-Gagnon, E., Desaulniers, G., Rousseau, L.-M., 2009. A branch-and-price-based large neighborhood search algorithm for the vehicle routing problem with time windows. Networks 54 (4), 190–204. doi:10.1002/net.20332.
Reyes, D., Savelsbergh, M., Toriello, A., 2017. Vehicle routing with roaming delivery locations. Transportation Research Part C: Emerging Technologies 80, 71–91. doi:10.1016/j.trc.2017.04.003.
Rochat, Y., Taillard, É.D., 1995. Probabilistic diversification and intensification in local search for vehicle routing. Journal of Heuristics 1 (1), 147–167. doi:10.1007/BF02430370.
Ropke, S., Pisinger, D., 2006. An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. Transportation Science 40 (4), 455–472. doi:10.1016/j.cor.2016.01.018.
Ropke, S., Pisinger, D., 2006. A unified heuristic for a large class of vehicle routing problems with backhauls. Eur J Oper Res 171 (3), 750–775. doi:10.1016/j.ejor.2004.09.004.
Santini, A., Ropke, S., Hvattum, L.M., 2018. A comparison of acceptance criteria for the adaptive large neighbourhood search metaheuristic. Journal of Heuristics 24 (5), 783–815. doi:10.1007/s10732-018-9377-x.
Savelsbergh, M.W., 1985. Local search in routing problems with time windows. Ann Oper Res 4 (1), 285–305. doi:10.1007/BF02022044.
Savelsbergh, M.W., 1992. The vehicle routing problem with time windows: minimizing route duration. ORSA Journal on Computing 4 (2), 146–154. doi:10.1287/ijoc.4.2.146.
Shaw, P., 1998. Using constraint programming and local search methods to solve vehicle routing problems. In: International Conference on Principles and Practice of Constraint Programming. Springer, pp. 417–431. doi:10.1007/3-540-49481-2_30.
Sitek, P., Wikarek, J., 2017. Capacitated vehicle routing problem with pick-up and alternative delivery (CVRPPAD): model and implementation using hybrid approach. Ann Oper Res 1–21. doi:10.1007/s10479-017-2722-x.
Solomon, M.M., Desrosiers, J., 1988. Time window constrained routing and scheduling problems. Transportation Science 22 (1), 1–13. doi:10.1287/trsc.22.1.1.
Souffriau, W., Vansteenwegen, P., Vanden Berghe, G., Van Oudheusden, D., 2013. The multiconstraint team orienteering problem with multiple time windows. Transportation Science 47 (1), 53–63. doi:10.1287/trsc.1110.0377.
Stützle, T., 2018. Heuristic Optimization.
Subramanian, A., Uchoa, E., Ochi, L.S., 2013. A hybrid algorithm for a class of vehicle routing problems. Computers & Operations Research 40 (10), 2519–2531. doi:10.1016/j.cor.2013.01.013.
Takada, Y., Hu, Y., Hashimoto, H., Yagiura, M., 2015. An iterated local search algorithm for the multi-vehicle covering tour problem. In: IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), pp. 1242–1246. doi:10.1109/IEEM.2015.7385846.
Tellez, O., Vercraene, S., Lehuédé, F., Péton, O., Monteiro, T., 2018. The fleet size and mix dial-a-ride problem with reconfigurable vehicle capacity. Transportation Research Part C: Emerging Technologies 91, 99–123. doi:10.1016/j.trc.2018.03.020.
Toth, P., Vigo, D., 2014. Vehicle Routing: Problems, Methods, and Applications. SIAM doi:10.1137/1.9781611973594.
Tricoire, F., Romauch, M., Doerner, K., Hartl, R., 2010. Heuristics for the multi-period orienteering problem with multiple time windows. Computers and Operations Research 37, 351–367. doi:10.1016/j.cor.2009.05.012.
Turkeš, R., Sörensen, K., Hvattum, L.M., Barrena, E., Chentli, H., Coelho, L., Dayarian, I., Grimault, A., Gullhav, A., Iris, Ç., Keskin, M., Kiefer, A., Lusby, R., Mauri, G., Monroy-Licht, M., Parragh, S., Riquelme-Rodríguez, J.-P., Santini, A., Santos Vinicius, G.M., Thomas, C., 2019. Meta-analysis of metaheuristics: Quantifying the effect of adaptiveness in adaptive large neighborhood search. Working Paper D/2019/1169/002. University of Antwerp. https://hdl.handle.net/10067/1636090151162165141
Vansteenwegen, P., Souffriau, W., Van Oudheusden, D., 2011. The orienteering problem: a survey. Eur J Oper Res 209 (1), 1–10. doi:10.1016/j.ejor.2016.04.059.
Vargas, L., Jozefowiez, N., Ngueveu, S.U., 2015. A selector operator-based adaptive large neighborhood search for the covering tour problem. In: International Conference on Learning and Intelligent Optimization. Springer, pp. 170–185. doi:10.1007/978-3-319-19084-6_16.
Vidal, T., Crainic, T.G., Gendreau, M., Prins, C., 2013. A hybrid genetic algorithm with adaptive diversity management for a large class of vehicle routing problems with time windows. Computers & Operations Research 40 (1), 475–489. doi:10.1016/j.cor.2012.07.018.
Wilcoxon, F., 1945. Individual comparisons by ranking methods. Biometrics bulletin 1 (6), 80–83. doi:10.2307/3001968.
Yıldırım, U.M., Çatay, B., 2015. An ant colony-based matheuristic approach for solving a class of vehicle routing problems. In: International Conference on Computational Logistics (ILCL). Springer, pp. 105–119.
Yuan, Y., Cattaruzza, D., Ogier, M., Semet, F., 2018. A branch-and-cut algorithm for the generalized traveling salesman problem with time windows. ROADEF 2018. Lorient, France
Yuan, Y., Cattaruzza, D., Ogier, M., Semet, F., Vigo, D., 2019. The generalized vehicle routing problem with time windows. VeRoLog 2019. Seville, Spain
Zhou, L., Baldacci, R., Vigo, D., Wang, X., 2018. A multi-depot two-echelon vehicle routing problem with delivery options arising in the last mile distribution. Eur J Oper Res 265 (2), 765–778. doi:10.1016/j.ejor.2017.08.011.